

From Archives to Algorithms: Tracing Trade and Pastoral Transitions in the Western Himalayas

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Computing and Human Sciences by Research

by

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CERTIFICATE

This is to certify that work presented in this thesis proposal titled ***From Archives to Algorithms: Tracing Trade and Pastoral Transitions in the Western Himalayas*** by *Hitesh Goel* has been carried out under my supervision and is not submitted elsewhere for a degree.

Date

Advisor: Dr. Aniket Alam

To Maa, Papa and Khushi

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Abstract

This thesis is situated in the intersection of History and Natural Language Processing, and examines the historical transformation of trade and pastoralism in the Western Himalayas, while also introducing *PastPaths*, a domain-adapted Natural Language Processing (NLP) pipeline developed to extract and visualize historical trade networks from unstructured textual archives. The Western Himalayan region, including present-day Kashmir, Ladakh, Himachal Pradesh, and parts of Tibet, has historically functioned as a dense zone of exchange, sustained by the movements of pastoralist communities and transregional trade in commodities such as wool, salt, and grain. Existing historiography has primarily framed the twentieth-century decline of pastoralism as a top-down consequence of colonial policies, often sidelining the agency of local communities. Through a close reading of archival records, census data, and ethnographic literature, this thesis offers a more nuanced account by highlighting how pastoralists strategically adapted to shifting political and economic contexts. These transitions included movement into agriculture, wage labor, and state employment.

From a computational perspective, the thesis addresses limitations in historical methodology by developing *PastPaths*, a pipeline that integrates OCR, Named Entity Recognition (NER), relation extraction, and map generation. The pipeline uses transformer-based models fine-tuned on manually annotated data from the region and is capable of extracting structured information about the trade of commodities. Our results show improved performance over baseline models and demonstrate the feasibility of large-scale trade reconstruction in data-scarce settings.

The thesis contributes to both historical and computational research. It reframes dominant narratives about pastoral decline by recovering voices of adaptation and choice, while also showcasing how domain-specific NLP tools can meaningfully support historical inquiry. More broadly, it advances the field of computational humanities by proposing an integrative workflow where historical questions guide model design, and computational outputs inform previous scholarship in the human sciences. This approach offers a framework for bridging disciplinary silos and generating new insights from underexplored textual archives.

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List of Related Publications

- [P1] Hitesh Goel, Easwar Balaramkrishnan, and Aniket Alam, **“PastPaths: Mapping Historical Trade From Texts”**, in *Turkey Computational Social Sciences*, 2024.
- [P2] Hitesh Goel, Aniket Alam, **“Structural Constraints and Historical Agency: The Political Economy of Pastoralism in the Western Himalayas Under British Rule”**, in *Indian Historical Review journal*, 2025 [Under Review].

Chapter 1

Introduction

1.1 Background and Motivation

1.1.1 Historical Background: Trade and Pastoralism in the Western Himalayas

The Western Himalayas, comprising present-day Kashmir, Jammu, Ladakh, parts of Gilgit, Hunza, and Baltistan, Himachal Pradesh, as well as regions of Uttarakhand and Western Tibet, have historically constituted a vital yet understudied ecological and commercial interface linking Central Asia, the Tibetan plateau, and the Indian subcontinent [1, 2]. Despite the region's formidable terrain and high altitudes, it sustained intricate and enduring circuits of transregional exchange. Traders' caravans and pastoralist herds transported wool, salt, borax, grain and other food, textiles, as well as a range of other commodities, across an inter-connected network of river valleys and mountain passes, connecting ecologically distinct zones and diverse political-economic formations [3–7]. Markets such as Leh and Gartok in western Tibet, Rampur (site of the Lavi fair) in the Sutlej valley, Srinagar in the Jhelum basin, Mandi in the Beas valley, and Hunza near the Pamirs functioned as nodal points where Himalayan and Tibetan traders and pastoralists, and merchants from Punjab and adjacent plains regions congregated to exchange goods and forge social relations [5].

Crucially, these networks were not maintained by sedentary agrarian societies alone. Transhumant pastoralists like the Gaddis, Gujjars, Bhotiyas, and Changpas were essential to the functioning of these economies [8, 9]. Their seasonal movements enabled ecological adaptation while supplying key trade commodities and providing logistical support through pack animals [10, 11]. The Himalayas' economic infrastructure was thus deeply shaped by mobile forms of life, rooted in flexible access to alpine pastures, reciprocal arrangements with agrarian communities, and participation in long-distance trade.

In secondary literature, we can find extensive exploration of how over the nineteenth and twentieth centuries, this intertwined system of trade and mobility underwent radical transformation [3, 12]. Colonial interventions, particularly forest reservations, land settlements, taxation reforms, and the re-routing of trade infrastructure, eroded the institutional basis of pastoralism [12, 13]. Simultaneously, shifts in global markets, such as the industrial reproduction of Kashmiri shawls in Europe, collapsed traditional demand for fine wool [14, 15]. This forced the shawl weavers and pastoralists in the region to look to

alternate professions to sustain their livelihoods. However, while structural constraints imposed by the colonial state played a significant role, in this thesis we show that the existing historiography has tended to overemphasize this aspect. It often portrays pastoral communities as passive victims of modernizing regimes, underplaying the role of local agency, internal decision-making, and adaptive transitions (Chapter 3).

1.1.2 Limitations in Current Methods of Historical Research

Despite the growing interest in trade and mobility in mountain regions, the historical reconstruction of these networks remains methodologically constrained. Much of the primary data like colonial gazetteers, ethnographies, revenue records, and travelogues, is textual, fragmented, and unevenly digitized [16, 17]. Scholars have traditionally relied on close reading and qualitative analysis to reconstruct commodity flows, route structures, and socio-economic transformations [18, 19]. While these methods have yielded valuable insights, they are difficult to scale. The lack of structured data and little interdisciplinary collaboration between historians and the computer scientists has meant that there have been limited efforts to map regional trade networks.

Moreover, even when digitized, these texts present challenges: they span multiple languages and scripts, are often plagued by OCR errors, and use inconsistent or archaic place names [20, 21]. As a result, extracting structured knowledge, such as which commodities were traded between which locations, is a labor-intensive process. There remains a pressing need for methodological innovation that can bridge the gap between rich historical narratives and the computational capabilities available today.

1.1.3 Opportunities in Computational Humanities

Recent advances in Natural Language Processing (NLP) offer promising tools to address some of the limitations mentioned above. Named Entity Recognition (NER), relation extraction, and domain-adaptive language modeling enable automated identification of people names, locations, commodities, and the interactions between them from unstructured texts [22–25]. While these tools have been successfully applied to classical literature, newspapers, and social media analysis, their use in the South Asian historical context, especially in the Himalayan region remains extremely limited.

This is not merely a technical gap, but a disciplinary one. There exists little integrated work that combines deep historical contextualization with domain-adapted computational models. Most NLP tools are developed on modern language corpora and fail to generalize to historical sources because of reasons like OCR errors and inconsistent spellings. Even fewer pipelines exist for extracting higher-order relationships such as trade routes or commodity flows, rather than merely recognizing entities. Furthermore, no open-source datasets or models currently exist for the Western Himalayas which can be used for building such pipelines, despite the region’s rich historical documentation.

The thesis addresses this gap by designing and implementing a computational model (Chapter 4), a domain-adapted NLP pipeline for historical text analysis. We use transformer-based models, fine-tuned on annotated data from the region, to extract information about the trade of commodities in various

regions from archival texts. In doing so, not only can our tool support historical inquiry into Himalayan trade and pastoralism but also contributes new resources to the field of computational humanities.

1.1.4 Interdisciplinary Disjunctures and the Need for Integration

At the core of this thesis is an attempt to bridge the methodological and epistemological divide between the human sciences and computer science. While historians and anthropologists have developed a sophisticated understanding of Himalayan ecologies and economies, they have largely worked with small-scale textual corpora. Conversely, NLP researchers have made remarkable progress in automated text analysis but often remain disconnected from the specific needs of historical scholarship, particularly in non-Western or multilingual contexts.

Few collaborative platforms exist that allow historians to frame questions in computationally tractable ways, or that enable computer scientists to develop tools grounded in domain-specific knowledge. As a result, computational methods are either underutilized or poorly aligned with actual historical questions. By combining critical historical analysis with customized NLP models, this thesis attempts a dialogic approach. It asks not only how computational tools can assist historical research, but also how historical problems and sources can push the boundaries of computational methods. In doing so, it situates itself within the growing field of computational humanities, which seeks not merely to digitize historical materials but to enable new forms of inquiry at the intersection of computation and history.

1.2 Research Questions and Contribution

This thesis brings together two distinct but complementary strands of research. On one hand, it engages with historical scholarship on trade and pastoralism in the Western Himalayas, particularly during the nineteenth and twentieth centuries. On the other, it examines how recent advances in computational methods, especially in Natural Language Processing (NLP), can be applied to historical corpora to extract structured data at scale. Our review has shown that while both domains are well-developed in their own right, there has been little interdisciplinary work that combines them in the specific context of the Western Himalayas.

As discussed in the previous section, from the humanities perspective, a major gap lies in the limited attention paid to pastoralist agency in the face of colonial and postcolonial transformations. Existing literature has largely attributed the decline of pastoralism to state interventions, often overlooking how local communities may have navigated, negotiated, or even actively shaped these changes. On the computational side, we find a lack of scalable tools and datasets tailored to this region. Moreover, existing NLP work on historical texts tends to focus on entity recognition alone, with relatively little emphasis on extracting relationships that could help reconstruct trade flows and mobility networks.

In light of these gaps, this thesis seeks to answer the following two interrelated research questions:

- *RQ1 (Human Sciences)*: How did trade and pastoralism evolve in the Western Himalayas during the nineteenth and twentieth centuries, and how did pastoral communities respond to shifting po-

litical and economic conditions? To what extent can we trace patterns of adaptation, negotiation, or decline that go beyond state-driven explanations?

- *RQ2 (Computer Science)*: How can computational methods be used to support historical inquiry, and how effective are they in extracting and mapping trade networks from historical textual sources?

To address these questions, the thesis makes contributions along three dimensions:

From a historical perspective, the thesis reinterprets the transformation of trade and pastoralism in the Western Himalayas by emphasizing the role of local agency. Through a close analysis of archival records, gazetteers, census data, and ethnographic literature, it identifies how pastoralist communities responded to colonial and postcolonial pressures not only through resistance but also through strategic adaptation. Rather than framing the decline of pastoralism as a straightforward outcome of structural constraints, the thesis recovers the diverse strategies through which these communities maintained mobility, sought new livelihoods, and negotiated with emerging institutions of power.

From a computational perspective, the thesis presents *PastPaths*, a domain-adapted NLP pipeline that extracts and visualizes historical place-commodity relationships from unstructured textual corpora. It advances existing work by moving beyond basic named entity recognition to include relation extraction and domain-adaptive fine-tuning using a curated dataset from the Western Himalayas. Our developed tool allows for the semi-automated construction of trade networks from historical texts, an advance that significantly improves the scalability and reproducibility of historical analysis in data-scarce contexts.

At an interdisciplinary level, the thesis contributes to the growing field of computational humanities by demonstrating how domain-specific historical knowledge and computational methods can be brought into a productive dialogue. The approach adopted here is not merely additive, but integrative: historical questions inform the design of computational tools, while the outputs of those tools feed back into historical interpretation. In doing so, the thesis offers a replicable model for interdisciplinary research, particularly in regions and topics where textual data is abundant but under-structured, and where traditional scholarly methods face limits of scale and consistency.

1.3 Structure of the Thesis

The remainder of this thesis is organized into three substantive chapters, each corresponding to a major component of the research questions laid out earlier:

Chapter 2: Literature Review provides a survey of existing work from both the human sciences and computer science. It first examines the historiography of trade and pastoralism in the Western Himalayas, identifying dominant themes and key debates. It then turns to recent developments

in computational methods, especially Natural Language Processing (NLP), and their application to historical text analysis. The chapter highlights methodological gaps in both disciplines and frames the need for an interdisciplinary approach.

Chapter 3: Structural Constraints and Historical Agency addresses RQ1 by reassessing the transformation of pastoralism in the Western Himalayas through archival records, census data, and ethnographic studies. While acknowledging the impact of colonial interventions, the chapter foregrounds pastoralist agency, mapping the varied strategies through which communities responded to changing economic and political conditions. It offers a regionally grounded and empirically supported narrative that complements and challenges existing literature.

Chapter 4: PastPaths - Mapping Historical Trade from Texts addresses RQ2 by presenting the design and implementation of *PastPaths*, a domain-adapted NLP pipeline developed to extract place-commodity relationships from unstructured historical texts. The chapter details the dataset creation and annotation process, the architecture of the NLP models, and the pipeline’s performance in named entity recognition and relation extraction tasks. It concludes by visualizing historical trade flows in the Western Himalayas, demonstrating the potential of computational methods to augment historical research.

Chapter 2

Literature Review

2.1 Introduction

This chapter surveys literature from two distinct but overlapping domains: the historiography of trade and pastoralism in the Western Himalayas, and recent developments in computational approaches to historical text analysis. It outlines key debates within each field, examining how historians have approached questions of mobility, exchange, and livelihood in the region, as well as how digital tools, particularly those from the field of Natural Language Processing (NLP), are increasingly being used to study historical sources at scale. By bringing these threads of research into conversation, the chapter highlights existing methodological gaps and sets the stage for an interdisciplinary approach to the study of trade and pastoralism in the Western Himalayas.

2.2 Historiography of the Western Himalayas

2.2.1 Broader Context

The Western Himalayas, which broadly include present-day Kashmir, Ladakh, Tibet, and Himachal Pradesh, have historically functioned as a critical zone for trade, mobility, and cultural exchange [1, 2]. Despite the forbidding terrain, long-distance trade routes threaded through high passes, linking South Asia with Central Asia and Tibet [3, 4]. Caravans of pack animals carried commodities ranging from luxurious cashmere wool to everyday salt and grain across these mountains [3]. Historians and anthropologists have increasingly highlighted how this region’s trade and mobility knit together diverse ecological zones and polities.

For instance, Andre Gunder Frank, in his book *ReOrient: Global Economy in the Asian Age*, reconstructs the global economy between 1400 and 1800, demonstrating how India and China were central to international trade [19]. Commodities such as silver, textiles, spices, salt, and wool circulated through dense, multilateral networks, leading Frank to argue that the so-called “Rise of the West” is a relatively recent phenomenon. Further, James Fisher’s ethnographic study, *Trans-Himalayan Traders* (1986), offers a more localized account of how trade in the Western Himalayas fostered both economic and

cultural exchange [26]. Based in the high-altitude village of Tarangpur in northwest Nepal, Fisher portrays the region as a crucial hinge between Tibetan Buddhist influences to the north and Hindu Nepali traditions to the south. The villagers participated in a “grain-salt-rice” exchange system by trading grain for rock salt brought by Bhotia traders from Tibet, and transporting the surplus salt southward in exchange for rice, a luxury in the highlands. These exchanges were not merely transactional, they were deeply embedded in social relations and cultural performance. As they moved across these trade networks, Tarangpurians strategically adopted different cultural personas, practising Mahayana Buddhism and speaking Tibetan in the north, while navigating caste structures and assuming Nepali identities in the south. While commodities moved fluidly across cultural borders, religious and ideological systems remained largely distinct. Fisher’s analysis highlights the adaptability of Himalayan communities, who acted not only as brokers of goods but also as mediators between contrasting cultural worlds. His work underscores how economic resilience and cultural pluralism were interwoven in the everyday lives of mountain societies.

2.2.2 Trade Routes and Circuits

The Western Himalayas were historically traversed by a dense web of trade routes that connected the Indian plains to Central Asia and Tibet [3, 5, 27]. Ladakh, in particular, served as a major transit hub, with trade routes branching southward to Kashmir, eastward to Lhasa in Tibet, and northward across the Karakoram Pass to Yarkand and Kashgar in Chinese Central Asia [1, 2, 28, 29]. These routes typically followed river valleys and were only accessible seasonally, opening in the summer months when snow melted. The Karakoram Pass route, rising to 5,575 meters, posed severe challenges to caravans and was notorious for the remains of traders and animals who had succumbed to the journey’s harsh conditions [30]. Yet, it remained central to Himalayan commerce, maintained jointly by the British and the Maharaja of Kashmir after 1870 [3, 31].

Trade routes were essential in sustaining the wool economy as they facilitated the movement and exchange of commodities across the Western Himalayas [32]. The primary route connecting Ladakh to Kashmir passed through Zoji La, a treacherous mountain pass that served as a conduit for Tibetan goods such as raw wool, salt, and other commodities [3]. Leh functioned as the central marketplace where traders from Tibet, Central Asia, and India converged. Tibetan traders brought wool, salt, and yak products to Leh, exchanging them for textiles, tea, and grains [3]. Beyond Leh, other important trade hubs included Lavi and Gartok, where merchants from Tibet, Ladakh, and Yarkand interacted with traders from Punjab and the Indian plains [33]. Wool and salt were transported via caravans to Kashmir, from where they were redistributed to other markets, while returning caravans carried grains, textiles, and metalware back into the Himalayas. Traders often used sheep and goats as pack animals, each carrying approximately twenty-four pounds (eleven kg) of goods, including salt and wool, across these mountain routes [33].

Leh’s bazaar, as contemporary observers noted, was a cosmopolitan marketplace bustling with activity during the summer months. The town served as an important link between South and Central

Asia, hosting permanent communities of Ladakhi Muslims and Buddhists, along with seasonal traders and pastoralists from across the region [2]. Farther east, routes from Ladakh to western Tibet passed via Pangong Lake and Demchok, linking markets like Gartok to Indian hill states [1]. Meanwhile, to the south, the city of Bushahr (in present-day Himachal Pradesh) controlled the Hindustan-Tibet Road, an ancient trading track revitalized under British rule. Starting in Shimla, this route traversed Kinnaur to reach the Tibetan border at Shipki La, where trade was jointly supervised by Bushahri and Tibetan officials [34–36]. The town of Rampur (Bushahr) hosted the annual Lavi Fair, which was a crucial marketplace where traders from Tibet, Ladakh, and the Indian plains met. Borax, pushum (fine goat wool) and churrus from Tibet were the dominant imports, while exports included raisins, pine nuts, tobacco, horseshoes, saddles, and firearms. Money was used for some transactions, but barter remained the primary mode of exchange, particularly for goods like salt and rice [5].

As noted earlier, long-distance trade in the Western Himalayas was typically conducted by caravans traversing segmented routes. Scholars such as Owen Lattimore have observed that few traders journeyed the entire distance from the Indian plains to Tibet or Central Asia [37]. Instead, goods commonly changed hands multiple times along the way. For instance, Ladakhi merchants would bring salt and pashmina to Leh, where these commodities were purchased by Kashmiri or Punjabi traders, who then carried them further downstream to markets in the plains. This relay-like system enabled goods to travel great distances, even as individual traders operated within shorter regional segments. Pack animals such as yak, dzo, ponies, mules, and camels were essential to this system, and though motor roads reached Leh in the mid-20th century, many traders continued to use traditional animal transport well into the post-colonial period [3, 38].

2.2.3 Major Trade Commodities and Flows

Now we shall take a look at the major commodities traded in the region. These goods moved in complementary directions: highland products such as wool, salt, borax, and pastoral by-products flowed downhill, while lowland goods like grains, textiles, tea, and manufactured items made their way up into the mountains. This multidirectional flow sustained the regional circuits from the 18th to the 20th century.

In the nineteenth century, the wool trade formed the cornerstone of the economies of Kashmir, Ladakh, and Tibet, underpinning an expansive network of trans-Himalayan commerce. Livestock was central to this economy, with sheep, goats, and cattle providing meat, milk, and wool [33, 39, 40]. Wool and its derivatives, especially pushum (pashmina wool), carpets, and the famed Kashmiri shawls were among the most sought-after commodities of the time and formed a vital part of the regional exchange economy [3]. The production of fine wool was primarily centered in the high-altitude pastures of Ladakh and western Tibet, where Changpa nomads raised pashmina goats and supplied raw wool, known locally as *asli tus* [3, 33]. Another prized material was *shah tus*, derived from the endangered Tibetan antelope, a rare commodity whose collection required the slaughter of the animal [14]. From these highland areas, traders facilitated the movement of wool to Kashmir, where it was processed in karkhanas (workshops)

into luxurious textiles. Kashmir's shawl industry held a global reputation, producing different varieties such as the *kani* (needle-woven), *amlakar* (embroidered), and *alwan* (plain) shawls [14,41]. These were valued across Central Asia, Europe, and South Asia alike.

Besides fine pashmina, the region also exported coarser sheep wool used in blankets, felts, and carpets. This “common wool” was traded in bulk from the Tibetan plateau and Ladakh to the hill states and plains. Kingdoms like Bushahr and Kullu served as intermediaries, importing wool for resale to the plains rather than local use [1, 40, 42]. As historian H.N. Kaul notes, these kingdoms competed to control wool imports for revenue generation, though only Kashmir possessed the technical expertise to convert it into high-value shawls. The scale of this commerce was vast: Moorcroft's early 19th-century data, analyzed by Rizvi, estimated that 120,000 to 240,000 lbs of pashm were imported into Kashmir annually, equating to the wool of hundreds of thousands of goats [43].

Salt and borax were also foundational to Himalayan trade. Much of the Western Himalayas and adjoining plains lacked local salt sources, making rock salt from Tibetan lakes such as Tso Kar and Tso Moriri especially prized [5, 7, 44]. Salt was harvested by Tibetan and Ladakhi pastoralists and transported by yak and sheep caravans to towns and valleys across Ladakh, Baltistan, and Himachal. British records from the 19th century describe extensive barter systems, whereby salt was exchanged for grain, rice, or cotton cloth [45]. Salt thus moved downhill while subsistence staples flowed uphill. In some remote valleys, salt functioned as currency. The trade often occurred at fairs such as Lavi in Rampur, where salt and brick tea from Tibet were exchanged for wheat, jaggery, and other goods [5]. Alongside salt, Tibetan nomads also collected borax, a mineral salt used in metalwork and medicine. Puga Valley in western Tibet was one of the key sources. As late as the 1870s, official records show 245 tons of borax and 365 tons of salt being imported annually through routes like Kumaon [46].

The high-altitude areas of Tibet and Ladakh were linked to the fertile plains of Punjab via trade in Himachal Pradesh. Tibetan wool, pashmina, borax, and salt were brought down into Himachal, while grains, tea, cotton textiles, and metals flowed in the opposite direction. Rampur, the capital of Bushahr, was the most important place in the region to which these commodities were brought [12]. As mentioned in the previous section, the Lavi fair held in the region was an important meeting point for traders across the region. Kinnaur, situated along the Sutlej Valley, also played a critical intermediary role in this trade, with entire villages actively engaged in commercial activities.

2.2.4 Decline Of Trade In the Region

While the wool trade had flourished under various ruling regimes in Kashmir, it also suffered repeated disruptions due to shifting taxation policies and external interventions [14]. During the Afghan rule (1752–1819), the *Dag-e-Shali* department was responsible for collecting shawls from weavers, which were then sold at prices significantly higher than their original value. This system later evolved into a tax collection mechanism from entire *karkhanas* (workshops). Under the subsequent *Qasur-i-Shali* system, duties were imposed on individual weavers. When the Sikhs took control of Kashmir (1819–1846), this taxation regime continued with slight modifications, including levies based on the number of shawls

produced [14]. There were periods when the tax burden became so severe that artisans were unable to sustain their livelihoods, prompting many to migrate to Punjab in search of better working conditions.

With the coming of British rule and the imposition of new policies, traditional wool trade networks came under further strain. The colonial administration promoted the Palampur route over the historically significant Sulej Valley corridor, undermining established Kashmiri and Tibetan trading interests. Moreover, the imposition of customs duties on caravans and the construction of regulatory checkpoints made participation in the wool trade increasingly expensive and bureaucratically cumbersome [3]. Compounding these logistical challenges, British manufacturers began to imitate Kashmiri shawls in European centres such as Norwich, Paisley, and Lyons. The invention of the Jacquard loom enabled mass production of shawl-like textiles at much lower costs, sharply reducing demand for authentic Kashmiri shawls [15].

The final blow to Kashmir's wool industry came with the outbreak of the Franco-Prussian War (1870–71), which disrupted European markets and caused a dramatic decline in demand. At the same time, the devastating famine of 1877–79 led to widespread starvation, including among Kashmiri artisans [14]. These twin calamities plunged the once-thriving shawl industry into irreversible decline. In the aftermath, many artisans shifted toward alternative occupations such as carpet weaving, work in state-run silk factories, or agriculture in rural areas of Kashmir [14, 47].

Although detailed records on the decline of external trade in Himachal Pradesh are limited, the downturn in Kashmir's trade, as outlined above, offers a useful comparative framework. Given the geographic proximity of regions such as Chamba to Kashmir, it is reasonable to infer that Himachal Pradesh experienced a similar economic contraction. Here too, colonial policies—particularly the restructuring of routes and the imposition of customs duties—likely played a significant role in disrupting established trade circuits and undermining traditional commercial practices.

2.2.5 The History Of Pastoralism In The Western Himalayas

In the Western Himalayas, pastoralist communities formed an integral part of the regional economy and ecology through the nineteenth century [8, 9, 13]. These groups often practiced transhumance, moving seasonally between highland and lowland pastures. For example, the Gaddis of Kangra (Himachal Pradesh) are renowned shepherds who traditionally spent summers grazing sheep and goats in the Chamba district and migrated south across the Dhauladhar range into Kangra's lower hills in the winter [10, 11]. Such cyclical migrations were central to the pastoralist way of living as it allowed them to exploit complementary environments, the lush mountain grass in the summer and the gentler climates during winter [48, 49]. Other pastoral groups in the region included the Gujjars, a nomadic buffalo-herding community who used to bring their buffaloes to the mountains during the hot months and retreated to the Shiwalik forests in winter, and the Bhotiyas of Kumaon and Garhwal, who kept flocks of sheep and goats not only for wool but also as pack animals, carrying goods across high passes between India and Tibet [13].

The Himalayan pastoralists played a key economic role by linking regions through the exchange of commodities. They often acted as intermediaries in trade networks, transporting goods over difficult terrain. For instance, the Bhotiyas engaged in the trans-Himalayan barter trade – they traded grains, jaggery, and other products from the Indian plains to Tibet in exchange for salt, borax, and wool [13]. In the northwest, Ladakh was central to the trade with Central Asia and Tibet, with the merchants exporting textiles, tea and spices from India while bringing back wool (especially pushum or cashmere wool), shawl wool and other valuables [50, 51]. The pashmina wool was primarily supplied by the Changpa nomads, who reared herds of the Changra goat, also known as the pashmina or cashmere goat. These goats were adapted to the harsh conditions of the Changthang plateau and produced an exceptionally soft undercoat during winter, which was carefully combed out in spring and transported to Ladakh's trading hubs, particularly Leh [3, 33]. Pastoralists also supplied regional markets with livestock products. Gaddi and Gujjar herders traded wool, ghee, hides, and meat to sedentary farming villages in return for grain and goods. For example, the Gaddi shepherds used to barter wool and carry salt on their sheep from salt mines in the Punjab plains up to hill markets [33]. In this way, mobile herders were pivotal in integrating the highland economy with the plains.

However, by the nineteenth and twentieth centuries, modern states and markets started to significantly affect the lives of the pastoralists by incorporating policies to settle nomadic herders and integrate them into the sedentary, taxable population [8]. In India, this was characterized by multiple policy changes under the colonial rule like increasing taxes on pastoral activities, limiting access to grazing areas by introducing forest and land settlements, and altering trade routes which significantly impacted transhumant activities in the Himalayas [13]. From the 1860s onward, colonial authorities demarcated large tracts of hill forests as 'Reserved Forests' under the Indian Forest Acts of 1865 and 1878 [13]. These laws curtailed traditional rights to graze, gather wood, or clear land inside forest reserves [12]. Unregulated grazing by pastoralists was now seen as incompatible with proper forest management and hence was tightly restricted by reducing the access to grazing lands [13]. Several scholars highlight how colonial policy deliberately shrank the space for nomadic pastoralism. The historian Chetan Singh notes that, whereas pre-colonial rulers saw value in mobile herders for revenue, the colonial state increasingly saw them as inconvenient and tried to regulate their movement [12]. British officials also imposed new taxes, such as grazing fees and transit duties on livestock caravans, that made pastoral livelihoods more costly. British land revenue settlements further required peasants and herders to pay taxes in cash, which pushed pastoralists to sell more of their products (wool, animals, butter) in markets to obtain money. Many pastoralists who once traded on barter (exchanging wool or milk for grain) had to commercialize their output to meet tax demands [6]. The cumulative effect was to erode the autonomy of pastoral groups and over time, these pressures led some nomadic groups to reduce their mobility and shift to alternate professions like agriculture.

By the late nineteenth and early twentieth centuries, the gradual hardening of borders in the Western Himalayas further disrupted the pastoral economy. Previously, pastoralists and traders moved freely across the region, maintaining networks that connected Ladakh, Tibet, Himachal Pradesh, and Central

Asia. However, colonial border policies, increasing state surveillance, and geopolitical shifts such as the Chinese annexation of Tibet (1950s) severely restricted trans-Himalayan trade [52–54]. Key trade routes, such as the Ladakh - Yarkand caravan trail and the Tibetan salt trade, were disrupted as customs checkpoints, permit requirements, and outright border closures made movement difficult. Ladakh went from a completely self-reliant economy to being dependent on supplies from the Indian government post Indian independence [55]. Furthermore, pastoralist communities, such as the Bhotiyas of Uttarakhand and the Changpa of Ladakh, who depended on cross-border trade like the Tibetan salt trade to sustain their livelihoods, found themselves cut off from traditional markets [13]. With the decline of these trade networks, pastoralists faced shrinking economic opportunities, forcing many to either reduce their herd sizes, settle in villages, or seek alternative sources of income. The loss of trans-regional trade accelerated the shift away from nomadic pastoralism, making these communities increasingly dependent on state support and local markets.

2.2.6 State of Research

The secondary literature offers a rich foundation for tracing the history of trade and pastoralism in the Western Himalayas throughout the nineteenth and twentieth centuries. As the preceding sections have shown, scholars have effectively demonstrated how colonial policies such as forest and land settlements, increased taxation, and the tightening of borders exerted mounting pressures on traditional pastoralist livelihoods. However, a significant gap in the literature remains: much of the existing scholarship attributes the decline of pastoralism primarily to state-led interventions, while overlooking the role of pastoralist agency and the possibility that some may have actively chosen to transition to alternative occupations.

In the following chapter, we first dive deeper into these existing arguments by examining them alongside primary sources such as gazetteers and census records, which allow us to substantiate and contextualise the claims made in the secondary literature. Moreover, to address the aforementioned gap, we will analyse post-independence census data in conjunction with anthropological studies and ethnographic surveys. This approach will enable a more nuanced understanding of the multiple factors that contributed to the decline of pastoralism. In doing so, we also hope to open new avenues for historically grounded research into pastoral communities in the Western Himalayas and their changing relationship with the state, market, and environment.

2.3 From Historiography to Computation

2.3.1 Limitations of current research methods in Human Sciences

In the previous section, we examined the extensive historical research on the Western Himalayas, which offers valuable insights into trade and pastoralism in the region. Scholars have drawn on travel-ogues, ethnographic accounts, and administrative records to reconstruct networks of mobility, patterns

of commodity exchange, and the effects of colonial policies on mountain communities. However, a persistent limitation remains: the primary sources are fragmented and largely qualitative. Efforts to extract structured data such as trade routes or commodity flows from historical documents have typically relied on manual annotation, a time-consuming process that lacks scalability, particularly when applied to large textual corpora [16–18].¹ This methodological constraint has long limited the scope of analysis in the field. Although rich textual evidence exists, the lack of scalable extraction tools has meant that historians must continue to rely on labor-intensive reading practices [56, 57]. As a result, researchers are often compelled to focus on a narrow set of commodities or a limited time frame when analysing patterns of exchange within a region.

2.3.2 Using NLP to Bridge the Gap

Recent advancements in Natural Language Processing (NLP) have opened up new possibilities for the automatic processing of historical texts, enabling the extraction of structured information at scale. These developments offer promising avenues for addressing the limitations faced by historians working with large, unstructured corpora. In this section, we examine a range of techniques and prior research efforts that have sought to bridge this gap between historical scholarship and computational methods.

2.3.2.1 Named Entity Recognition and Domain Adaptation in Historical Texts

Named Entity Recognition (NER) is the automated identification of entities such as people, places, and organizations, and serves as a foundational step in turning unstructured historical texts into structured data [23, 58, 59]. In historical research, NER can help retrieve key actors, locations, and spatial references, thus offering valuable insights into trade networks, social structures, and patterns of mobility [22, 23]. Early approaches to NER in historical texts were rule-based, relying on hand-crafted grammars and lists of known names [60]. However, these methods often struggled with the variability and inconsistency of historical language [58]. The development of statistical sequence models, particularly Conditional Random Fields (CRFs), introduced greater flexibility and improved performance, as long as annotated training data was available [61]. By the mid-2010s, neural network-based models such as BiLSTM-CRF architectures became the state-of-the-art, significantly improving NER accuracy on modern texts and showing promising results on historical corpora as well [62–65].

In recent years, the emergence of transformer-based language models such as BERT and RoBERTa have brought significant advances in Named Entity Recognition [66–68]. These models use self-attention mechanisms [69] and are pretrained on large text corpora, allowing them to capture detailed contextual information [70, 71]. When fine-tuned for NER, they achieve high scores and have become the standard architecture in many NLP systems [72]. Applied to historical texts, transformer models have outperformed earlier neural approaches, particularly in areas like 19th-century newspapers and medieval charters [58, 73]. However, one major challenge remains: most pretrained models are built

¹<https://www.digitalhumanities.org/dhq/vol15/4/000574/000574.html>

using modern language data, which makes them less effective for historical sources [74]. Historical texts often contain old vocabulary, inconsistent spellings, unusual punctuation, and references to entities that no longer exist. These differences between modern and historical language reduce the accuracy of off-the-shelf models when applied directly to historical corpora [75, 76].

To address these challenges, researchers have developed domain-specific strategies to improve NER performance on historical texts [77–79]. One common technique is text normalization, which involves converting archaic or inconsistent spellings into their modern equivalents [80, 81]. This approach is especially useful in languages that have undergone significant spelling reforms, such as French, or that display high orthographic variation, as seen in medieval Latin. Studies have shown that normalization can significantly improve NER accuracy, particularly when combined with syntactic preprocessing in highly inflected languages [82, 83].

Another effective strategy for adapting machine learning models to historical texts is domain-adaptive pretraining. Instead of using language models trained only on modern data, researchers continue pretraining these models on historical corpora to better capture the linguistic characteristics of the past [84, 85]. This process, also known as adaptive language modeling, helps the model learn the vocabulary, grammar, and usage patterns specific to a historical period or genre [86–88]. Manjavacas and Fonteyn (2021) compared two approaches: continuing pretraining on historical English using a modern BERT model, and training a new model from scratch on historical text [75]. They found that while training from scratch led to better results in tasks like part-of-speech tagging and NER, it also required much more data and computational power. Continued pretraining, by contrast, offered a more efficient alternative and still led to strong performance gains [75]. This approach has seen success in many specialized domains, such as biomedical and legal NLP, and is increasingly being applied to historical language [89]. Domain-adaptive models have been trained for ancient Greek, medieval Latin, and biblical texts, resulting in improved performance across tagging, entity recognition, and syntactic parsing [90, 91]. Notably, this approach helps us capture detailed context from the historical documents and to a large extent solves the problems of ambiguous place names or other entities.

In recent times, we have seen multiple attempts at integrating NER to better understand historical corpora and to extract relevant information from them. A notable example is the Edinburgh Geoparser, a tool that automatically recognizes place names and resolves them to geographic coordinates using gazetteers [92]. The tool has been successfully applied to 18th- and 19th-century documents, identifying locations across large corpora and laying the groundwork for historical GIS applications. Another widely used platform is Recogito, developed by the Pelagios project, which provides a semi-automatic workflow for annotating place names in both texts and images [93]. Recogito allows historians to link toponyms to gazetteer URIs and generate spatial visualizations of place references. Through crowdsourced annotation campaigns, over 120,000 place-name occurrences from classical and medieval sources have been identified and tens of thousands manually verified. These tools highlight how using NER for identifying geographic entities can transform narrative geography into structured, mappable data.

However, two key limitations remain in many of the existing NER-based projects. First, many of these projects continue to rely on rule-based approaches, which often fail to generalize across diverse historical corpora and languages, and do not incorporate the state-of-the-art NLP techniques discussed above to enhance their performance. Second, while NER is essential for identifying entities in text, it alone is not sufficient for capturing the higher-level relationships between them. For instance, to reconstruct historical trade routes, it is not enough to identify place names and commodities individually, we also need to extract the relationships between them, such as which goods were traded between which locations. This points to the need for additional techniques, such as relation extraction, that can model these more complex connections within historical texts.

2.3.2.2 Relation Extraction

As we have mentioned, historians are generally interested in extracting higher order relations between the named entities in a text. Understanding who traded with whom, which goods moved between regions, or how individuals were socially or economically connected requires the extraction of relational information from unstructured narratives. This task, known as relation extraction (RE), is a key challenge in information extraction and has seen considerable progress over the past decade [94, 95]. Traditional RE approaches relied on pattern-based rules or statistical classifiers to identify relationships between pairs of entities [96–98]. The rise of neural models in the 2010s, including convolutional and recurrent neural networks, improved the ability to extract semantic relationships from text by learning patterns across sequences of words [99, 100]. However, historical texts present unique challenges: relationships are often implied rather than explicitly stated, and resolving pronouns, coreferences, or vague descriptors (e.g., “the caravan,” “the company”) is essential for identifying the entities involved in a given interaction.

Recent advancements in large language models (LLMs) have opened new possibilities for automating relation extraction in historical corpora. For example, recent studies have come out that have LLMs like GPT-3 for relation extraction [24, 25]. The researchers found that GPT-3 generated more meaningful and context-aware relations than traditional semantic role labeling systems [101]. Unlike earlier pipelines, which relied heavily on syntactic parsing, the generative transformer model was able to infer connections even when relations were implicit or embedded in emotionally complex narrative structures. This reflects the broader strength of LLMs: they encode broad semantic knowledge and can adapt to noisy or loosely structured text, making them well-suited to the ambiguous and fragmentary nature of many historical sources.

2.3.2.3 OCR and Text Digitization

A foundational requirement for applying NLP methods to historical sources is the availability of machine-readable text. Much of the historical record, however, remains embedded in scanned books, manuscripts, and inscriptions. Optical Character Recognition (OCR) and Handwritten Text Recognition (HTR) are thus critical for converting digitized archives into analyzable formats [58, 102–104]. Despite

considerable technological advances, OCR performance on historical documents continues to suffer from significant limitations due to factors such as archaic typefaces, degraded page quality, and obsolete spelling conventions [105, 106]. As a result, the output often contains substantial errors, making OCR a persistent bottleneck for historical text analysis.

The effects of OCR noise on downstream NLP tasks are well-documented. Studies have shown that even modest declines in OCR accuracy can result in large drops in the performance of information extraction models [20, 21]. Van Strien et al. (2020), for example, found that F1 scores for person-name recognition fell from 87% to 63% when comparing clean text with noisy OCR output from historical newspapers [107]. Similarly, Hamdi et al. (2019) demonstrated that a BiLSTM-CRF-based NER system suffered substantial performance losses when confronted with OCR-induced spelling variations, especially for rare or domain-specific entities. Such errors, though seemingly minor (e.g., misreading “John” as “Joim” or “6” as “G”), often propagate through the processing pipeline, misleading both entity recognition and syntactic parsing modules [108].

Recent advances in Large Language Model (LLM)-based approaches have demonstrated state-of-the-art performance in a variety of OCR-related tasks, particularly when applied to challenging or noisy inputs [109, 110]. Unlike traditional OCR pipelines that rely on fixed character templates or rule-based post-processing, LLMs can leverage contextual understanding to correct transcription errors, infer missing or partially recognized tokens, and adapt to stylistic variations across time periods [111]. These capabilities make LLM-based methods particularly well-suited for processing archival materials, where domain-specific linguistic patterns and poor scan quality have historically posed significant barriers to reliable text extraction [112].

2.3.3 State of Research

This section has outlined a range of recent NLP techniques that have been applied to historical research, demonstrating their potential for processing and analyzing large volumes of unstructured textual data. However, a significant gap remains: very few of these interdisciplinary approaches have been applied to the context of the Western Himalayas. As a result, there is a severe lack of ready-to-use datasets and computational tools for analyzing travelogues, administrative records, and other primary sources from the region at scale.

Another key limitation, as discussed earlier, is the scarcity of research focused on extracting higher-order relationships between entities which is crucial for visualizing and interpreting complex historical trade networks. Most existing projects have concentrated on entity recognition in isolation, without linking these entities to model interactions such as commodity exchanges or trade routes.

This thesis addresses these gaps by proposing a novel approach aimed at building the foundational resources and tools necessary for large-scale computational analysis of historical trade in the Western Himalayas. In the final chapter, we present the design and implementation of a custom NLP pipeline developed for this purpose, which includes domain-adapted models and relation extraction techniques tailored to historical sources from the region.

Chapter 3

Structural Constraints and Historical Agency: The Political Economy of Pastoralism in the Western Himalayas Under British Rule

3.1 Introduction

In this chapter we shall re-evaluate the decline of pastoralism in the Western Himalayas, moving beyond the prevailing narrative that attributes this transformation solely to colonial intervention. While British policies such as taxation, land enclosures, and forest reservations undoubtedly imposed structural constraints on pastoral mobility [9, 13, 113], this chapter foregrounds the agency of pastoralist communities in shaping their own economic transitions. Drawing on archival sources and ethnographic studies, it highlights how groups such as the Gaddis of Kangra and Chamba strategically adapted to changing socio-economic conditions. Rather than being passive victims of state imposed pressures, many pastoralists diversified their livelihoods, integrating farming, wage labour, and government employment into their economic repertoires. Developments such as the expansion of government jobs [114, 115], increasing access to formal education [115, 116], and market driven changes further contributed to this shift, making traditional transhumance less viable and, for many younger generations, less desirable.

Pastoralism, integral to the economic and social fabric of the Western Himalayas, particularly in Himachal Pradesh, was historically sustained through transhumant herding practices [3, 12]. These enabled pastoral communities to exploit complementary ecological zones across seasons and maintain trade and subsistence networks across altitudes [3, 12, 117]. However, the visible decline of pastoralism through the twentieth century has drawn substantial scholarly attention. Much of the existing literature attributes this decline to external forces: colonial taxation, forest and land settlements, trade route disruptions, and a broader shift toward sedentary agriculture [6, 12, 115]. While these structural explanations are well supported by historical evidence, but they may not fully account for the complex social dynamics that accompanied the transformation of pastoral life.

As discussed in our literature review (§ 2.2.5), trade and pastoralism were deeply interlinked in the Western Himalayas during the nineteenth and twentieth centuries. While colonial policies exerted considerable pressure on these systems, we must also ask: was the decline of pastoralism entirely imposed from above, or did pastoralists themselves participate in and shape these transformations. This chapter addresses this question in detail. In the sections that follow, we examine colonial and post-

colonial records such as gazetteers, census data, and land revenue reports, alongside anthropological and ethnographic research. In doing so, we aim to substantiate existing claims about the impact of state intervention, while also investigating the extent to which pastoral communities themselves actively pursued alternative livelihoods.

We argue that the transition away from pastoralism cannot be understood solely in terms of state coercion or policy pressure. It also reflected internal motivations — economic, social, and cultural — that led many pastoralists to adopt occupations such as settled agriculture, wage labour, or government service. A close reading of census records and village surveys reveals evidence of this shift occurring as early as the mid-twentieth century. By shifting focus from structural constraints to historical agency, our work calls for a reassessment of colonial records and oral histories to better capture the complex motivations behind pastoralist transformations. Ultimately, it seeks to offer a more in-depth understanding of the social and economic change in mountain societies where we attempt to understand the dynamics of the interplay between external pressures and internal choices in shaping pastoralist futures.

3.2 The Kashmir Context And Comparison

Kashmir presents a well documented case for understanding the evolution and decline of Himalayan trade economies, particularly in the context of the wool industry. Its rich archival and secondary sources allow for a detailed analysis of how colonial interventions, market shifts, and internal restructuring transformed a once thriving commercial hub. This case also serves as a comparative baseline for understanding similar changes in regions like Himachal Pradesh.

As discussed in the literature review, Kashmir held a near monopoly over the processing and weaving of pashmina, and its shawl industry was among the most prestigious in the world (§ 2.2.3). The word “shawl,” derived from the Persian word *shall*, originally referred to a fine wool garment worn over the upper body [41]. Over the eighteenth and nineteenth centuries, Kashmiri shawls gained international renown for their craftsmanship, material quality, and intricate designs [14]. Artisans in *karkhanas* (workshops) produced various styles, including *kani* (needle woven) shawls made with *tujji* (small wooden needles), *amlakar* (embroidered) shawls, and *alwan* (plain) shawls, using fine wool such as *asli tus* and the extremely rare *shah tus*, obtained from pashmina goats and Tibetan antelopes, respectively [14,41].

However, this prestigious industry began to falter by the mid-nineteenth century. European textile centres such as Paisley and Norwich began mass producing imitation shawls using the Jacquard loom, drastically reducing production costs [15]. While a Kashmiri shawl cost between seventy to one hundred euros, a Paisley imitation could be purchased for just £12 [47]. This technological and economic competition was accompanied by fiscal exploitation under Dogra rule. For instance, Maharaja Gulab Singh’s administration imposed a blanket tax of Rs. 47-8 on each shawl weaver, irrespective of their actual income, thereby increasing the financial burden on artisan families and reducing profitability [47].

The combined effect of increased taxation and external market competition led to a steep decline in exports. Historical records show that between 1850–51 and 1860, the value of shawl exports to Europe plummeted from £171,000 to £51,000 [47]. By 1905, exports of Kashmiri shawls had virtually collapsed, registering a mere Rs. 1,000 [5]. Once a cornerstone of Kashmir's economy, the shawl industry could no longer sustain the same levels of employment or revenue, compelling many artisans to abandon their traditional crafts.

In response to this economic downturn, the state, with British encouragement, began promoting sericulture as an alternative to the failing wool economy. Kashmir's climate, altitude, and abundance of mulberry trees made it an ideal site for silk production [118, 119]. New silk reeling facilities were established in Srinagar, and sericulture rapidly became an important sector of the regional economy. The transition was facilitated by the fact that silk production, like wool weaving, involved various stages of labour from silkworm rearing to thread spinning and weaving, thus absorbing displaced workers from the shawl industry [14]. Trade records illustrate this shift clearly: the value of silk exports increased from Rs. 7,000 in 1897–98 to Rs. 13.6 lakhs in 1902–03, and reached nearly Rs. 21 lakhs by 1904–05 [118].

These shifts also reshaped Kashmir's broader import-export dynamics. Until the late nineteenth century, Kashmir was relatively self-sufficient, importing modest quantities of salt, sugar, tobacco, tea, metals, and petroleum. However, with the rise of new industries and integration into colonial trade networks, imports increased rapidly in the early twentieth century (Table 3.1). On the export side, total trade surged from Rs. 53.3 lakhs in 1892–93 to Rs. 192 lakhs in 1904–05, though the nature of these exports had shifted dramatically. The wool trade, which once made up nearly half of all exports—particularly through the valuable pushum (fine goat wool) trade—had dwindled to insignificance by 1905 [5]. Meanwhile, the share of manufactured goods using wool as a base material increased, further signaling the industry's shift toward modern industrial processes (Table 3.2).

Despite this economic transition, the cultural significance of wool production, especially shawl weaving, persisted. Artisans in Kashmir continued to produce carpets and woollen textiles for local use and for small scale exports. The practice of weaving remained closely tied to individual and community identity, and was strongly associated with Kashmiri Sufi traditions and notions of craftsmanship [5, 14]. Even as large scale commercial production declined, many artisans chose to remain in the profession due to its symbolic and cultural value.

3.3 Trade in Himachal Pradesh

Trade in Himachal Pradesh during the colonial period was deeply embedded within the larger trans-Himalayan economy, linking the region with Tibet, Ladakh, Punjab, and Central Asia [12]. The economy was characterised by a combination of agriculture, pastoralism, and trade. Commodities such as wool, salt, grains, borax, and textiles moved across ecological and political boundaries, supported by a network of commercial centres including Rampur, Kinnaur, Mandi, Kangra Proper, Lahaul, and

Commodity	1892–93	1902–03	1904–05
Piece goods:			
European	76,688	3,472,934	3,592,556
Indian	21,572	960,186	786,485
Metals:			
Brass and Copper	133,143	142,410	99,888
Iron	131,270	492,020	119,567
Salt	483,293	869,761	932,601
Sugar:			
Refined	482,584	921,872	828,675
Unrefined	115,433	247,686	343,761
Tea:			
Indian	173,730	357,638	522,871
Foreign	2,030	1,360	3,615
Tobacco	101,253	232,302	391,960
Petroleum	56,112	81,795	184,164

Table 3.1 Kashmir imports (in Rupees), 1892–93 to 1904–05 [120].

Spiti [5, 12]. These towns and valleys functioned as key hubs for goods flowing between the highlands and the plains.

The structure of society in Himachal during this period significantly shaped both economic activity and trade. Social organisation was primarily clan-based, with groups such as the Khash forming patrilineal, exogamous lineages that lived in polyandrous joint families. Governance was conducted through the *khumri*, a secular council of male clan representatives responsible for revenue collection, dispute resolution, and resource management. Religious and political life were closely intertwined, with local deities and their functionaries mediating social order and seasonal cycles of pastoralism and agriculture [6]. Unlike Kashmir, where feudal landholding hierarchies were more prominent, Himachal’s social system was decentralised and based on shared access to land and resources. Authority was not concentrated under a landed nobility but rather negotiated within kin-based institutions that regulated rights under the symbolic authority of local deities [6].

In addition to facilitating material exchanges, trade in the region fostered complex forms of cultural interaction. As noted in the literature review, James F. Fisher’s study of Himalayan trade in Tarangpur highlights how traders adopted multiple cultural identities depending on where they were operating (§ 2.2.1). Though Fisher’s work focuses on Nepal, it provides a useful lens to interpret similar dynamics in Himachal Pradesh. Given the region’s geographic location between the Tibetan plateau and the Indian plains, it is likely that traders from areas such as Kinnaur and Lahaul engaged in similar forms of cultural duality, embracing Buddhist customs when trading in Tibet and aligning with Hindu practices when dealing with traders from Punjab. Although direct evidence is limited, this interpretive framework provides insight into how trade facilitated not just the movement of goods, but also of identities and affiliations.

Commodity	1892–93	1902–03	1904–05
Drugs, not intoxicating	160,625	506,192	578,425
Dyes	274,826	95,533	48,023
Fruits	238,683	458,702	748,883
Hides	186,594	297,617	798,160
Skins	139,386	343,350	563,065
Ghi	1,650,172	2,005,849	3,002,974
Linseed	1,335	655,674	73,952
Wool:			
Manufactured piece-goods	591,439	751,365	1,075,047
Shawls	219,275	35,600	1,000

Table 3.2 Kashmir exports (in Rupees), 1892–93 to 1904–05 [120].

Pastoralism was tightly interwoven with these trade networks. Kinnaur, in particular, was known for its transhumant herding systems, where pastoralists migrated seasonally between high altitude and low altitude pastures [117]. Sheep were especially valuable for their wool, which was used in local textile production as well as for trade. Goats served as essential pack animals, transporting items such as salt, grains, and textiles [12]. The Gaddis of Kangra and Chamba were among the most prominent pastoralist communities in the region. Their wealth was measured in livestock rather than land, and they performed cyclical migrations as they grazed their flocks in alpine pastures during the summer and descending into the lower hills during the winter months [6]. Their herds provided wool, dairy products, and meat, and played a vital role in regional markets and barter systems.

Trade fairs and marketplaces were the epicentres of this commercial and cultural activity. Among the most significant was the Lavi Fair, held annually in Rampur. It brought together traders from Tibet, Ladakh, and the Indian plains. Tibetan imports included borax, pushum (fine goat wool), and churru, while local exports ranged from raisins and pine nuts to firearms and saddles. Barter remained the dominant mode of exchange, particularly for basic goods like salt and rice, though monetary transactions also occurred [5]. Trade records from 1840 show that white pushum was the most valuable import at the Lavi Fair, worth Rs. 89,600, followed by wool at Rs. 8,971 and black pushum at Rs. 4,832 (Table 3.3).

Other key trading centres included Mandi, Palampur, and Kangra, which were closely linked to the Punjab market. Mandi, in particular, emerged as an important transit hub in the Beas Valley after the British oversaw the development of roads connecting Pathankot and Hoshiarpur to Mandi, which were later extended to Manali [50]. Palampur, in the Kangra district, was also a prominent commercial hub, known for its annual fair that attracted merchants from Yarkand, Kulu, Lahaul, and various districts of Punjab, including Amritsar, Jalandhar, Hoshiarpur, and Ambala [121]. In 1871, traders from Yarkand primarily brought horses, charas, carpets, woolen goods, and silver to the Palampur fair, while merchants from Himachal traded pashmina, grains, vegetables, English piece goods, and jewelry. The total value of these imports stood at Rs. 3,24,501. However, by 1873, this figure had sharply declined to Rs. 65,000

Commodity	1837	1838	1839	1840
White Pushum:				
Quantity (Katcha Maund)	1,185	1,481	1,463	1,400
Rate (Rs per 2 Pucca Seers)	3½	4–1	6–1	8–1
Value (Rupees)	33,180	47,392	70,224	89,600
Black Pushum:				
Quantity (Katcha Maund)	175	190	119	151
Rate (Rs per 2 Pucca Seers)	1½	2–1	3–1	4–1
Value (Rupees)	2,425	3,040	2,856	4,832
Wool:				
Quantity (Katcha Maund)	1,092	784	867	1,570
Rate (Rupees)	8	8	7	8
Value (Rupees)	5,460	3,920	3,935	8,971
Borax:				
Quantity (Katcha Maund)	394	566	678	505
Rate (Seers per Rupee)	13	16	13	16
Value (Rupees)	1,189	1,415	818	621
Churrus:				
Quantity (Katcha Maund)	215	236	241	571
Rate (Seers per Rupee)	8¼	8¼	8¼	8½
Value (Rupees)	10,750	8,496	10,249	2,394

Table 3.3 Exports from Tibet to Rampur (Lavi Fair), 1837 to 1840 [5].

as the fair's significance waned, with traders shifting their focus to emerging commercial centres such as Shimla [121].

This decline in traditional trading centres like Palampur reflected the broader economic reconfiguration under British rule. The colonial administration introduced new commercial circuits that redirected trade away from older routes and local fairs, concentrating economic activity in emerging hubs like Shimla. By integrating Himachal's markets into the broader colonial economy, the British sought to monetize transactions and move away from the barter based networks that had historically structured trade. As a result, Shimla became a key market node, symbolizing the shift from an exchange economy to one shaped by British economic priorities [12]. To reinforce this transformation, the colonial government invested in infrastructural projects that reshaped trade networks across the region. The Hindustan-Tibet Road, developed along the Sutlej Valley, improved connectivity between Rampur, Kinnaur, and Tibet, strengthening British control over trans-Himalayan commerce. Other major routes included the Pathankot - Mandi - Kulu road, the Hoshiarpur - Mandi road, and the Ropar - Mandi road, which linked Himachal Pradesh to key trading centres in Punjab [50]. These infrastructural expansions facilitated the movement of goods but also reinforced British economic dominance, ensuring that trade flows aligned with colonial interests rather than local commercial traditions.

Wool was a central commodity in the trade networks of Himachal Pradesh, particularly in Kinnaur, Spiti, Lahaul, and Kulu. Shepherds in these regions transported large bundles of raw wool to trading

hubs such as Rampur and Mandi, where it was further sold to merchants from Punjab and British India [50]. Among the various types of wool produced in the region, the finest came from wild goats, which played a commercially significant role. As these goats moved through thorny shrubs and rocky terrain, they naturally shed the softest fibres of their undercoat. Local shepherds collected this wool, which is also known as shah tush or asli tush, as we have already mentioned in a previous section, and was highly valued for its exceptional quality. A distinct community of wool gatherers and shepherds, known as the Khang-pas, relied entirely on this vocation, making the collection and trade of fine wool central to their livelihood [50, 122]. Wool exported from Tibet was also carried by Lahauli traders, who took cash into Tibet during the summer months to purchase wool, which was then transported south to Kulu and Amritsar for further distribution. Besides wool, other commodities such as honey, ghi, cumin, apricots, and yak tails were exported down the Sutlej Valley to Punjab. Trade with Tibet also brought in salt, borax, and dried fruits, many of which were re-exported to markets in British India [123].

Tehsil	1850	1855	1868	1881
Kangra	154,599	181,086	211,165	218,588
Nurpur	139,252	148,993	127,368	105,244
Debra	87,099	91,833	126,294	121,423
Hamirpur	203,577	214,875	180,133	176,609
Kangra proper	584,427	636,766	644,959	621,865
Kulu and Plach	—	82,189	89,914	100,259
Lahaul	—	2,535	6,970	6,860
Spiti	—	2,087	3,039	2,863

Table 3.4 Population in the Kangra district from 1850 to 1881 [124].

The British imposed taxes on shepherd communities, including the Gaddis of Kangra, to regulate wool production. In 1914, the tax rate on sheep and goats was raised to one anna per goat and nine pies per sheep, placing an economic burden on pastoralists who depended on large herds for their livelihoods [125]. Another significant shift under British rule was the promotion of sericulture. Silk production emerged as an alternative economic activity, particularly in Nurpur and Triloknath, offering employment to those affected by the decline of wool based industries. Table 3.4 illustrates the population changes in the district between 1850 and 1881. Notably, the population of Kangra tehsil grew from 154,599 in 1850 to 218,588 in 1881, largely due to the expansion of tea cultivation. The cultivated area also increased significantly from 435,940 acres in 1868 to 717,360 acres in 1881, reflecting a broader shift away from the wool trade in the latter half of the nineteenth century [124].

While specific data on the decline of external trade in Himachal Pradesh is limited, the trajectory closely parallels that of neighbouring Kashmir, as also mentioned in the literature review (§ 2.2.4). The geographical and economic similarities suggest that Himachal, too, faced a comparable downturn driven largely by colonial interventions.

3.4 Pastoralism And Its Decline

As discussed in the previous section, pastoralism was a crucial part of Himachal Pradesh's economy and livelihood patterns. However, by the late 19th and early 20th centuries, colonial policies began reshaping these traditional practices through increased taxation, forest regulations, restrictions on grazing, and a push towards sedentary agriculture. British administrators, concerned with balancing commercial forestry, agricultural expansion, and pastoral land use, introduced measures that gradually curtailed the mobility of pastoralists. These policies altered traditional grazing rights, imposed new taxes, and enforced stricter controls over forest lands, significantly impacting transhumant communities such as the Gaddis, Gujjars, and Kinnauris. In this section, we first map out pastoralism in Himachal Pradesh, detailing its economic significance, migratory patterns of the pastoralists, and integration with trade and agriculture, and then examine how colonial policies contributed to the decline of nomadic herding in the Western Himalayas.

3.4.1 Pastoralism in Himachal Pradesh

Pastoralism is particularly prevalent in the high altitude regions of Kinnaur, Lahaul, Spiti and Chamba. The region's rugged terrain, limited arable land, and extreme seasonal variations necessitated a transhumant lifestyle, where herders moved seasonally between high altitude summer pastures and lower winter grazing lands. This practice allowed them to efficiently utilize natural resources and facilitated the integration of pastoralism with trade and agriculture [9, 12]. Communities such as the Gaddis, Gujjars, and Kinnauris played a prominent role in sustaining the pastoral economy. Sheep and goats were the primary livestock, providing wool for local use and trade, while buffaloes, owned largely by Gujjars, were valued for their dairy products. The pastoral economy was deeply intertwined with trade, as livestock not only provided commodities like wool and dairy but also served as pack animals for the transport of salt, grains, and textiles across the Himalayas [5].

The seasonal migration of pastoralists was a defining feature of the region's economy, as it allowed the pastoralists to use a range of ecological areas at different times of the year. In Kinnaur, Lahaul, and Spiti, livestock movements followed a strict cycle, with flocks grazing in lowland pastures during winter and shifting to high altitude pastures during summer. Families divided their animals into 'kim zet', those kept permanently in the village and 'arung zet', those moved across pastures to ensure optimal resource use [113, 126]. A major hub for pastoral exchange was Gartok, a summer market where pastoralists and traders from Central Asia, Ladakh, and Tibet congregated to exchange goods. Sheep and goats were key trade assets, not only for their wool but also for their role as transport animals in trans-Himalayan commerce [113].

Among the best known transhumant pastoralists of the region were the Gaddis of Kangra and Chamba. Their extensive migrations allowed them to access pastures across multiple climatic zones. Many Gaddi herders held hereditary grazing rights (warisi) over specific pastures, particularly in Chamba, Kangra, and Bangahal. Some pastures were held collectively, while others were leased from local

rulers [12, 113, 124]. Warisi rights ensured that Gaddi flocks could access grazing areas seasonally, with obligations such as penning sheep in local fields to fertilize crops. Similarly, the Gujjars followed a migratory pattern, herding buffaloes across the lower hills and specializing in dairy production. Unlike the Gaddis, who relied primarily on sheep and goats, Gujjars engaged in milk and ghee trade, often supplying settled agricultural communities [6]. Pastoralists played a crucial role in the local economies of the regions where they migrated. For the peasantry, they provided much needed manure to fertilize the land. Additionally, the pastoralists of Lahaul and Kinnaur used their flocks as pack animals, aiding in the transportation of goods and facilitating trade between India and Tibet. In Kulu, Lahaul, and Spiti, pastoralism was the primary livelihood for many communities [113].

Numbers from the Kangra Gazetteer of 1924 [125] show that Gaddi herders annually brought down more than 250,000 goats and sheep from Chamba and Kulu to graze in the lower hills during the winter months. Gujjars also migrated with their buffalo herds from Chamba, contributing to dairy production in the region. While British policies would later restrict some aspects of pastoralism, the initial decades of colonial rule saw an expansion of sheep and goat populations as demand for wool increased. By 1912, the number of sheep and goats in Kulu had increased by twenty four percent over 1891, reflecting a growing emphasis on wool production [123]. Data shows us that within Kulu Proper, the number of sheep and goats in 1909 was 95,565 which increased to 1,18,554 by 1912. The number of oxen in the region also showed a slight increase of 4.16 percent increasing from 34,392 in 1891 to 35,823 in 1912 [12].

For the initial years of British rule in the region, the pre-existing taxation structures on sheep and goats were maintained, with Kangra shepherds paying a standard rate of one anna per head per annum [124]. This allowed pastoralists to continue their seasonal migrations with minimal interference in the earlier colonial period. British administrators later formalized grazing taxation through the *timi* (grazing tax), which in the early nineteenth century had been paid in kind through wool or livestock [123]. Despite the new tax regulations, grazing claims remained largely unchanged during the early British period, as the colonial administration continued to recognize traditional pastoral rights, including the warisi rights of the Gaddis and the sowana rights of the Gujjars over certain forests and pastures [125]. Table 3.5 gives details on the number of agricultural livestock in the Kangra district over the years. As we can see there was a significant jump in the number of livestock between 1894 and 1909, with goats showing the most significant increase from 62,404 in 1894 to 445,560 in 1909.

Livestock Type	1894	1909	1914
Bullocks and Cows	348,697	440,360	439,488
Buffaloes	88,678	117,919	119,272
Young stock	137,414	174,042	172,613
Sheep	110,181	172,564	169,484
Goats	62,404	445,560	473,642

Table 3.5 Livestock in Kangra district from 1894 to 1914 [125].

Pastoralism in Himachal Pradesh remained a vital economic and social institution throughout the nineteenth and early twentieth centuries. The transhumant lifestyle of Gaddis, Gujjars, and Kinnauris enabled them to sustain trade, supply wool, and support agrarian economies through manure and dairy production. In the next section, we will examine how colonial policies began restricting pastoralism through forest regulations, grazing limits, and increased taxation, ultimately shaping the decline of traditional herding practices in the Western Himalayas.

3.4.2 The Decline Of Pastoralism

The decline of pastoralism in Himachal Pradesh has been a subject of extensive scholarly study. Existing literature attributes this decline to a combination of colonial policies, increased taxation, forest reservations, trade disruptions, and the broader shift toward sedentary agriculture [12, 13]. This section will look into the major explanations offered by previous studies while integrating primary sources from revenue records, gazetteers, and administrative reports to provide empirical support for these arguments. While these structural and external factors undoubtedly shaped pastoralism's trajectory, the agency of pastoralists like Gaddis in adapting to these changes, and themselves wanting to shift to other professions and occupations, has not been given the importance it deserves in existing literature. This question will be taken up in the next section.

One of the earliest disruptions to pastoralism came with the British assertion of control over forests and grazing lands. Before colonial rule, pastoralists had free access to common pastures, but under the Forest Act of 1865 and its expansion in 1878, large tracts of land were designated as reserved forests. Further rules were enacted in 1891 as the British looked to control the grazing access of the pastoralists like the Gaddis [12, 125]. This shift was driven by British demand for timber, particularly for railway expansion, leading to the commercialization of forests at the expense of pastoral communities as unrestricted grazing was seen as a threat to forest regeneration [5]. The forest reservation policies in Kulu, which culminated in the demarcation of protected areas between 1865 and 1897, marked a turning point. By 1917, nearly sixty percent of Kulu's total forest area was classified as reserved or demarcated forest, leaving pastoralists with increasingly restricted grazing lands [12].

Range	Reserved Forests		I Class Demarcated		II Class Demarcated		Undemarcated (acres)
	Number	Area (acres)	Number	Area (acres)	Number	Area (acres)	
Kulu	11	3,692	46	10,912	66	113,985	—
Rupi	18	19,821	48	35,041	22	429,788	—
Inner Saraj	9	11,357	53	11,005	46	77,814	—
Outer Saraj	7	5,009	36	16,037	34	42,586	—
Total	45	39,879	182	72,995	179	664,173	538,781

Table 3.6 Area and number of different classes of forests in Kulu, Rupī, and Saraj ranges [123].

The forest settlements of the late nineteenth and early twentieth centuries formalized four types of forest categories: Reserved Forests, fully controlled by the government with no local rights to graz-

ing or wood collection; Demarcated Protected Forests, where limited grazing and timber rights were allowed under strict regulation; Undemarcated Protected Forests, where villagers retained some access but under increasing restrictions; and Unclassed Forests, which were theoretically open for community use but were gradually brought under state control [125]. These systematic demarcations were particularly extensive in Kangra, Nurpur, and Palampur, leading to large portions of village commons being converted into state controlled reserves [125]. Table 3.6 gives details on the area of different classes of forest in 1917 in the Kulu subdivision. As a result of this classification of forest land into reserved and demarcated, many traditional summer pastures for Gaddis, who relied heavily on sheep and goat herding, and Gujjars, predominantly buffalo herders, were lost, forcing them to find alternative routes, reduce herd sizes, or abandon transhumance altogether [50].

The situation was further worsened by the introduction of monetized grazing taxes, which transformed pastoralism into a financially burdensome occupation. Earlier, pastoralists either paid minimal dues or had reciprocal relationships with agrarian communities, exchanging wool, dairy, and manure for grazing access. However, the British administration increasingly sought to extract direct revenue from pastoralists by introducing fixed grazing fees. The 1891 taxation reforms marked the beginning of this process, as the government began distinguishing between local flocks and foreign flocks, imposing additional levies on pastoralists grazing outside their native kothis or ‘revenue districts’ [12]. This shift must be understood in the context of pre-colonial practices, where a lot of the pastoral lands functioned as commons, i.e. shared grazing spaces accessed seasonally by multiple transhumant communities. Under colonial rule, however, these commons came under increasing regulation. The administration introduced tiered grazing taxes that penalized non-local users. For instance, the 1883–84 Kangra Gazetteer records that while local flocks were taxed at Rs. 1-9-0 per hundred sheep, the rate for “foreign” sheep rose to Rs. 3-2-0 [124]. These escalating costs severely eroded the economic viability of pastoralism, forcing many herders to either reduce flock sizes, sell their livestock, or transition into settled agriculture [125].

As grazing lands shrank and taxes increased, the livestock population in Himachal Pradesh began to decline, while cultivated land expanded as communities were pushed toward settled farming. Between 1911 and 1945, the number of sheep and goats saw a sharp decline. Table 3.7 presents a comparison of the number of sheep and goats in the Kulu subdivision between 1911 and 1945. As we can see, the overall decline in the number of sheep and goats in the Kulu region is approximately twenty percent, with the Rupi region experiencing the most significant decrease at 42.63 percent. Inner Saraj and Outer Saraj also recorded substantial declines of around twenty three percent. On the other hand, the decline in the population of oxen was relatively much smaller at around ten percent, reflecting a transition from pastoralism to plough based agriculture [12]. Table 8 shows the change in the cultivated area in the Kulu subdivision between 1911 and 1945. As we can see, the cultivated area increases in all the regions and its overall increase is more than five percent. This further shows how colonial policies actively displaced traditional herding practices in favour of farming [12]. The impact of these policies by the British was most visible in Nurpur, which used to be the centre of shawl manufacture in the nineteenth century. If we look at the variation in the population of the region as given in the Kangra Gazetteer of 1924 [125],

Waziri	Year	Sheep	Goats	% Change (Goats)
Kulu Proper	1911	77,369	41,186	
	1945	77,284	31,874	-7.92
Rupi	1911	25,563	19,682	
	1945	16,295	9,661	-42.63
Inner Saraj	1911	15,658	10,766	
	1945	11,592	8,821	-22.74
Outer Saraj	1911	29,848	19,472	
	1945	23,411	14,195	-23.75
Lahaul	1911	22,313	6,385	
	1945	20,549	5,830	-8.09
Spiti	1911	2,237	2,536	
	1945	2,076	1,990	-14.81
Average Decline (%)				-19.37

Table 3.7 Comparison of sheep and goat numbers in the Kulu subdivision between 1911 and 1945 [12].

we see that the population was consistently declining in the twentieth century, as it first declined by 2.2 percent between 1901 and 1911 and then by 4.6 percent between 1911 and 1921. Another region that faced massive population decline was the Kangra tehsil, where the population declined by over five percent between 1911 and 1921. . Beyond taxation and land encroachment, pastoralists also suffered due to disruptions in the wool trade. The abolition of customs duties through British treaties favoured new trade routes such as the Palampur - Leh - Kashmir corridor, which diverted commerce away from the traditional Sutlej Valley trade networks [127]. As a result, pastoralists who had once relied on selling wool and livestock in fairs like Lavi, Mandi, and Shimla found their economic model under threat [6]. As we have noted in a previous section, imports to the Lavi Fair had declined significantly by 1873, further illustrating the economic pressures faced by these pastoral communities. At the same time, the construction of roads, particularly the Hindustan-Tibet Road, introduced new obstacles for transhumant groups. While roads facilitated commercial trade, they also created new conflicts over grazing land and subjected pastoralists to increasing state surveillance [113]. Herders were required to register their livestock at fixed checkpoints, such as Wangtu and Sarahan in Bushahr, adding layers of bureaucracy and taxation. Meanwhile, state imposed boundaries on grazing zones made it impossible for pastoralists to follow traditional migration routes, significantly altering the patterns of Himalayan transhumance [113].

Recent reports suggest that with pastoralism becoming increasingly unviable, many herding communities were forced into sedentary agriculture or wage labour. By the late nineteenth century, the Gaddis of Kangra and Chamba had already begun losing access to traditional pastures, compelling many to settle in fixed locations [50]. In Lahaul, Spiti and Kinnaur, increasing state control over grazing lands forced pastoralists to abandon transhumance altogether. This shift not only affected their economic stability but also led to an erosion of the cultural identity of these pastoralists, which is also reflected in the declining number of Gaddi pastoralists in the twentieth century [50].

3.5 Internal Push To Alternate Professions

Until now we have seen how the decline of pastoralism in Himachal Pradesh has been attributed to external factors such as colonial taxation policies, forest restrictions, and trade disruptions, and how the primary records corroborate these arguments to a large extent. However, we wish to question the notion that these external factors sufficiently explain the decline of pastoralism in the twentieth century. Historical records also seem to suggest that the shift away from transhumance was not solely imposed by the government authorities but also driven by the choices of the pastoralists themselves. Over time, many of them actively sought alternative livelihoods, moving towards agriculture, government employment, and other occupations that offered greater economic stability and social status.

Rather than just being passive victims of colonial policies, pastoralists like the Gaddis appear to have made strategic adaptations to the realities of a shifting political economy. Census data and administrative records indicate that even as restrictions on pastoralism intensified, many Gaddis were already diversifying their livelihoods, engaging in mixed agro-pastoral practices, trade, and other skilled professions. Education, access to new forms of employment, and increased interaction with settled communities further facilitated this transition. This section examines how internal economic motivations, social mobility, and the pursuit of alternative livelihoods contributed to the decline of transhumance among the pastoralists, complementing the well documented external pressures imposed by the colonial state.

The Village Survey of Brahmaur in the Census of India 1961 [128] provides crucial evidence of a semi-pastoral, semi-agricultural economic structure amongst the Gaddi pastoralists. It notes that, while sheep and goat herding remained a dominant livelihood, a significant number of people were engaged in agriculture, weaving, silversmithy, and blacksmithy. The report further mentions that the yearly migration to Kangra and other places was on the decline as Gaddis increasingly stayed in their villages during winter months, reflecting an early transition away from full transhumance to a more sedentary lifestyle. The survey attributes this change to new avenues of livelihood becoming available with the opening of roads, which suggests that mobility was no longer a necessity for economic survival.

Data on occupational patterns and household income in Brahmaur village shows that Gaddis who practiced agriculture alongside sheep breeding earned higher incomes than those solely reliant on pastoralism. The highest income group (above Rs. 100 per month) in the village included 106 out of 146 households, with a significant portion of them engaged in agriculture, labour, and sheep breeding. Government service also emerged as a preferred profession, with forty seven families reporting government employment, with a majority of them in the highest income group, further reinforcing the idea that alternative livelihoods were becoming more desirable and accessible. The Village Survey of Maingal [129] presents similar findings, noting that while Muslim Gujjars primarily depended on buffalo rearing, the Gaddi Brahmins of the village practiced a mix of agriculture and goat and sheep rearing, indicating that some pastoral groups were already integrating agricultural activities into their economy. Additionally, the survey recorded cases where Gaddi households engaged in weaving, spinning, and basket making, demonstrating a broader occupational base beyond just pastoralism.

A consistent theme in recent anthropological research is that education has played a critical role in dissuading younger Gaddis from continuing transhumance [115, 116, 130] . The 1961 Village Survey of Brahmaur shows a higher literacy rate (twenty nine percent) compared to the district average of nine percent, suggesting that education was already gaining importance among Gaddi communities in the early years of independence. Interviews conducted in the 2019 survey of Gaddi transhumance further confirm that formal education has reduced younger generations’ willingness to engage in pastoralism. Among surveyed respondents in the study, forty nine percent cited education as a key reason for leaving herding, with many opting for government or private sector jobs instead. This pattern aligns with trends observed in the 1961 census, where a considerable number of Gaddi families had already transitioned to government employment. Similarly, Saberwal (1999) [115] and Malhotra et al. (2022) [114] found that Gaddi families increasingly invested in government jobs, particularly in the police and military, as these occupations offered greater financial security compared to the uncertainties of transhumance. The transition to government service aligns with mid-twentieth century administrative records, which document an increase in state sponsored employment for tribal populations in Himachal Pradesh [128, 129].

This shift among the Gaddis is consistent with broader social science research, which has extensively documented how education fosters occupational mobility, often drawing individuals away from traditional livelihoods toward more lucrative, stable employment. Studies from Nepal [131] illustrate how education has become synonymous with escaping physically demanding occupations such as farming, with families explicitly investing in schooling to secure non-manual, white collar employment. Similarly, research on caste-based artisan communities [132] shows that formal education provides an alternative to hereditary trades, allowing individuals to pursue government or private sector jobs instead. Leavy and Smith (2010) further highlight that in sub-Saharan Africa, rural youth overwhelmingly associate education with improved economic prospects outside of agriculture, leading them to seek alternative professions when educational opportunities are available. The Gaddi experience thus fits within a larger global pattern where schooling increases access to non-traditional employment, reinforcing a broader shift away from labour intensive, historically embedded occupations.

Waziri	Year	Cultivated Area (acres)	Acres Increase	% Increase
Kulu Proper	1911	29,150		
	1945	30,954	+1,804	+6.18
Saraj	1911	38,394		
	1945	39,499	+1,105	
Rupi	1911	11,526		+2.87
	1945	12,685	+1,159	
Total	1911	79,070		+10.05
	1945	83,138	+4,068	+5.14

Table 3.8 Change in cultivated area in the Kulu subdivision between 1911 and 1945 [12].

As we have already seen in the previous section, the livestock numbers were on a steep decline in the first part of the twentieth century. Table 7 shows that between 1911 and 1945, the number of sheep and goats declined by approximately twenty percent, with some areas like Rupri recording a decline of as much as 42.63 percent. This reduction in herd sizes coincided with an increase in cultivated land. In Kulu, the cultivated area expanded by over five percent between 1911 and 1945 (Table 3.8). More recent studies confirm that Gaddi pastoralists used surplus income from herding to invest in land purchases, as agriculture offers a more stable and predictable economic return [11, 115]. This suggests that the shift to agriculture was not simply because of colonial policies but also because the Gaddis found the profession to be more economically rewarding, perhaps also less physically demanding and giving them a better social status. This expansion of cultivated land by the Gaddis continued after independence, as evidenced by the increase in the total area sown with rabi crops in Brahmaur village from thirty acres in 1956 to 62 acres in 1962 [128].

A significant challenge to continuing pastoralism has also been the breakdown of joint family systems and the increasing preference for nuclear households. Multiple studies highlight that the transition from joint to nuclear family structures has made it difficult to sustain labour sharing models traditionally required for transhumance [114, 130]. The 2019 survey of Gaddi pastoralists revealed that only twenty one out of forty six respondents were still engaged in full-time pastoralism, with the rest working as farmers, shopkeepers, or daily wage labourers. This supports the idea that even during British rule, labour constraints within families may have contributed to the shift away from pastoralism.

Beyond economic factors, urbanization and political participation have reshaped Gaddi aspirations. Pal (1961) [129], notes that new roads, government presence, and increased political awareness were influencing Gaddi livelihoods. Over time, many Gaddis actively sought political representation, culminating in their reclassification as a Scheduled Tribe in 1956, which further facilitated access to government jobs and education [133]. Recent studies confirm that Gaddi youth today view pastoralism as backward and less socially prestigious [115, 116]. This changing perception suggests that even in the early-twentieth century, pastoralism may have been increasingly viewed as a less desirable occupation relative to emerging alternatives. However, livelihood choices and the ability to respond to such shifts were not uniform across the community. Internal hierarchies, particularly along the lines of gender have long shaped the distribution of opportunities and expressions of agency. The community is largely male-dominated, with herding responsibilities traditionally assigned to men. Even among younger generations, studies have noted a clear gender disparity in access to opportunities: boys are more likely to be given access to formal education and mobility, while girls are expected to remain within domestic spheres or help with agricultural labour [13]. This unequal distribution of labor and access definitely influences how different members of the community are able to respond to changing economic conditions and thus the agency exercised by the community is not evenly distributed [13]. It is also important to note that the Gaddi community does not exhibit internal caste divisions. While class-based stratification has begun to emerge in more recent decades, this development lies outside the temporal scope of our study and has therefore not been addressed in detail.

While British policies played a dominant role in restricting pastoralism, evidence from primary sources and later anthropological research suggests that Gaddis were not mere victims of the policies of the colonial and developmental state, but were also partly active in transitioning away from traditional transhumance. This shift was driven by education, government employment, agricultural expansion, land ownership, labour shortages, and changing social aspirations. Although the available historical records do not directly capture the internal decision making of Gaddi pastoralists under British rule, census reports and economic data suggest that these transitions were already underway by the mid-twentieth century, and thus can be traced back to at least the early-twentieth century. Later ethnographic studies provide a compelling framework to interpret this earlier shift, demonstrating that pastoralists were not simply reacting to colonial policies, but actively reshaping their economic and social lives in response to new opportunities.

3.6 Conclusion

The historiography on the Western Himalayas, particularly in the past half century, typically shows the mountain people as mere victims of colonialism and modernity. This trend is even more stark in the historiography on pastoralism and forests. While the depredations of colonialism (and of the post-colonial developmental State) are important to record, this often leads to a narrative which eclipses the historical agency of the people who inhabit mountain societies. Pastoralists in the Western Himalayas were deeply affected by colonial policies. This paper has examined the wool trade in the Western Himalayas under British rule, highlighting its economic significance and the factors that contributed to its transformation. While colonial policies such as taxation, land restrictions, and shifting trade routes played a crucial role in the decline of pastoralism, our analysis underscores the need to re-examine census reports and other primary sources from the colonial era to better capture the agency of pastoralist communities in shaping their own economic transitions. The Gaddi pastoralists of Himachal Pradesh, in particular, did not merely react to external pressures but actively sought alternative livelihoods in agriculture, government employment, and trade. By integrating primary sources with existing literature, this study challenges the prevailing narrative that attributes the decline of pastoralism solely to colonial interventions. Instead, it argues that internal economic motivations, increased literacy, and social aspirations among pastoralists played a key role in this shift. A more thorough engagement with colonial records and oral history is necessary to fully understand how pastoralists navigated these transitions, ensuring that historical narratives do not overlook their role as active agents of change.

Chapter 4

PastPaths: Mapping Historical Trade From Texts

4.1 Introduction

Historical trade networks offer critical insights into the economic, cultural, and political dynamics of a region. These networks reveal how goods, people, and ideas moved across complex geographies and political boundaries, shaping societies in both direct and subtle ways. In the context of the Western Himalayas which is a region marked by high altitude passes, linguistic and cultural diversity, and strategic transregional connections, the study of historical trade holds particular importance. From wool caravans traversing mountain routes to salt and grain exchanges across valleys, the region’s archival record testifies to a vibrant and interdependent economic landscape that linked the Indian subcontinent with Tibet, Central Asia, and beyond.

As discussed in the preceding chapters, historians have made significant contributions to understanding this trade system through meticulous archival research, ethnography, and analysis of colonial records (§ 2.2,3). However, these methods often remain labor-intensive and limited in scale [16–18]. The primary sources available such as gazetteers, travel accounts, administrative reports, are largely unstructured and fragmented, making it difficult to extract comprehensive, structured insights without substantial manual effort [56, 57]. This has constrained historical inquiry, particularly when it comes to mapping commodity flows or reconstructing long-distance trade routes at scale [134–136].

In recent years, computational techniques, especially those grounded in Natural Language Processing (NLP), have emerged as powerful tools for addressing these limitations. By automating tasks such as Named Entity Recognition (NER), Relation Extraction, and geoparsing, NLP offers the ability to process large corpora of historical texts systematically and with greater precision [137–139]. While such methods have seen application in domains like social media analysis, they remain underutilized in the context of South Asian mountain societies, where a lack of curated datasets, inconsistent vocabulary within texts, introduction of noise because of digitisation of the documents present unique challenges [140–142].

This chapter presents a novel end-to-end computational pipeline developed to extract, structure, and visualize trade-related information from historical texts concerning the Western Himalayas in the nineteenth and twentieth centuries. Specifically, the pipeline enables users to identify and visualize which

places are involved in the trade of a given commodity, as described within a particular historical document. It integrates OCR-based preprocessing, domain-adapted Named Entity Recognition (NER), and large language model (LLM)-based relation extraction, followed by the visualisation of the trade on a cartographic map. Our aim is to demonstrate how computational methods can scale historical analysis while also revealing relational and spatial patterns that are difficult to discern through manual close reading alone. We envision our work as a stepping stone to other software tools in the future that would help historians in analysing trade networks at scale.

The chapter is structured as follows. We begin by describing the dataset creation process, a foundational and time intensive stage, given that the performance of downstream models is highly dependent on the quality and relevance of the training data. We then outline each component of the pipeline, from OCR and text preprocessing to NER, relation extraction, geocoding, and visualization. Evaluation results and a sample reconstruction of historical trade networks are presented to illustrate the system’s practical utility. We conclude with a discussion of key methodological challenges, limitations, and directions for future work. By bridging approaches from history and computer science, this chapter proposes a scalable and reproducible framework for computationally studying historical trade in the Western Himalayas.

4.2 Data

4.2.1 Dataset Creation

Named Entity Recognition (NER) is a central component of our computational pipeline, as it enables the automatic identification of key entities such as place names, commodities, and people within historical texts [22, 143]. This task is crucial for extracting structured information from unstructured sources and lays the foundation for subsequent steps such as relation extraction and geographic visualization. However, the performance of any NER system is highly dependent on the quality and relevance of the dataset it is trained on. Models trained on inappropriate or mismatched data tend to misclassify or miss entities altogether, especially when applied to a historical and domain-specific context like ours [144–146].

Off-the-shelf NER datasets, such as the widely used CoNLL-2003 [147] corpus or OntoNotes 5.0 [148], are typically constructed from modern texts like Reuters news articles, web data, or conversational transcripts. These datasets are optimized for identifying contemporary entities such as “New Delhi,” “Barcelona,” or “Lionel Messi” in well formatted, modern English prose. As a result, models trained on these datasets often struggle when applied to historical documents, which may feature archaic spelling, colonial-era toponyms, and non-standardized syntax. For instance, a model trained on modern newswire data will fail to recognize historically significant but now obscure place names like “Kullu,” “Chumba,” or “Zaskar” as entities, or confuse commodities like “pashm” or “borax” for common nouns rather than trade relevant terms.

Given this mismatch, and in the absence of any existing annotated corpora tailored to the study of trade in the Western Himalayas, we constructed a custom dataset specifically for our task. We manually curated a corpus of 22 historical documents from a larger corpus of 181 documents collected from the digital archive PAHAR. These documents primarily comprise travelogues, surveys and official reports authored by British officers during the colonial period, dating from the early 19th century to the mid-20th century.¹ The selection was curated to ensure both temporal and spatial coverage: documents were chosen to span the entire 19th century and to collectively represent different parts of the Western Himalayan region. This allowed the training data to reflect the diversity of place names, routes, and trade-related observations across time and space in the region, making it ideal for a domain-adapted NER model. The annotation process involved tagging location names, commodities, and other relevant entities within these texts, forming the foundation for a specialized tool that can support historical trade analysis at scale.

4.2.2 Annotation Process

The annotation process focused on three core entity types that are central to the analysis of historical trade networks: **Geographical Entities**, **People**, and **Commodities**. These categories were selected to capture the spatial, social, and economic dimensions of trade as documented in the historical texts. Annotation was carried out using the *Prodigy* tool [149], which offers a streamlined interface for iterative annotation and active learning, and integrates effectively with the *spaCy* NLP framework. This allowed us to refine the entity recognition model in real time and maintain high annotation efficiency.

Two trained annotators independently labeled the corpus. The annotation guidelines were developed in an iterative manner, informed by both linguistic analysis and subject matter expertise in South Asian history. A reconciliation step followed the initial round of annotation to resolve disagreements through discussion and cross-referencing. The resulting dataset was evaluated for consistency using inter-annotator agreement metrics, yielding a Cohen’s Kappa score of 0.83 and an F1 score of 0.96 indicating strong agreement and high quality annotations.

The *Geographical Entities* category included a wide range of spatial references as found in the historical texts. These encompassed habitations such as Haridwar, Srinagar, and Kedarnath; rivers and lakes like the Yamuna, Bhagirathi, and Mansarobar; as well as mountains, peaks, and high passes such as Shipki La. Other kinds of landscape references like forests, roads, and sites of local importance (e.g., Indra Dhara, Sansar Dhara, and Masret Chawra) were also annotated under this category. This granularity was important for capturing the varied geographies through which trade and mobility took place in the Western Himalayas.

The *People* category captured the wide social diversity present in the documents. It included private individuals named in the sources (e.g., Puki Sonar, Harblaram), but also collective identities such as Paharis, Tartars, and Butias. Political and official titles such as Raja, Lieutenant, Jemadar, and Shista

¹Link to the documents used for annotation:https://drive.google.com/drive/folders/1GscLT7FLvkFkCwkX_UZVAdETNhoHsTP?usp=sharing

Tapah were annotated to reflect the role of state and administrative actors. Additionally, social and caste identities, which are crucial to understanding patterns of participation in trade, were included, examples being Rajputs, Gosains, Sannyasis, and Jogis. Religious figures like Ram-Rae, Daulet Reo Sindel’a, and Dhoulal Gir were also tagged under this category when mentioned as active agents in social or economic contexts. Annotation in this category was particularly sensitive to the fluidity between caste, occupation, and ethnic identity in the colonial vocabulary.

The *Commodities* category was designed to cover both staple and luxury goods that circulated through the trade networks. Past works [5, 6] were referenced to come up with an exhaustive list of such commodities and these included minerals and salts such as rock salt and borax; various types of wool including sheep wool, yak wool, and finer grades like pushm white and pushm black; as well as botanical and edible commodities like cannabis, asafoetida, figs, almonds, pistachio nuts, prunes, raisins, and pomegranates. Several varieties of cloth were annotated, such as muslin, mashru, sarcenet, and local weaves like pattus and kharua. In cases where commodities were embedded in complex phrases or inferred from context rather than explicitly stated, annotators applied the guidelines to ensure consistent tagging. These detailed annotations provide a solid foundation for training models that can extract commodity-flow relationships from historical texts with both precision and nuance.

Overall, the annotation process was tailored to the challenges of historical language, regional specificity, and the fragmented nature of the source material. The resulting dataset constitutes a domain-specific, high quality corpus suitable for training a Named Entity Recognition model capable of operating effectively in the underexplored context of nineteenth and twentieth-century trade in the Western Himalayas.

4.2.3 Dataset Characteristics

As mentioned previously, in total 22 different documents were annotated. A document wise split was performed to obtain the train and test sets, 17 documents and their annotated entities were used for training and the rest of the 5 documents were used as the test set. Overall, the corpus looks like this:

Data	Train Set	Test Set
Sentences	54,071	15,348
Tokens	507,798	145,422
Geographical Entities	10,368	3,282
People	4,029	1,168
Commodities	907	242

Table 4.1 Statistics of the training and test sets.

The dataset spans a diverse range of registers, reflecting colonial administrative prose, ethnographic observation, and transliterated vernacular expressions.²

²Link to the annotations: <https://drive.google.com/drive/folders/1KhqKMd9h9j150QUCg8QIpAXznI9gvv7L?usp=sharing>

The table below summarizes the annotated entity types, their scope, and representative examples drawn from the corpus:

Entity Type	Description	Examples
Geographical Entities	Geographical identifiers specific to the Western Himalayas, including settlements, valleys, rivers, and passes.	Garhwal, Kumaon, Simla, Pindari Pass
People	Individuals or groups mentioned in the documents, often denoting merchants, colonial officials, nomadic tribes, or ethnic communities.	Sayid, Gujur, Kuchis, Bhotia traders
Commodities	Trade goods referenced in the texts, including food items, textiles, raw materials, and luxury products.	Wheat, rice, shawl, salt, opium

This dataset forms the foundation for the subsequent NLP tasks described in the pipeline. It is also intended as a reusable resource for future research at the intersection of computational history and South Asian studies.

4.3 Methodology

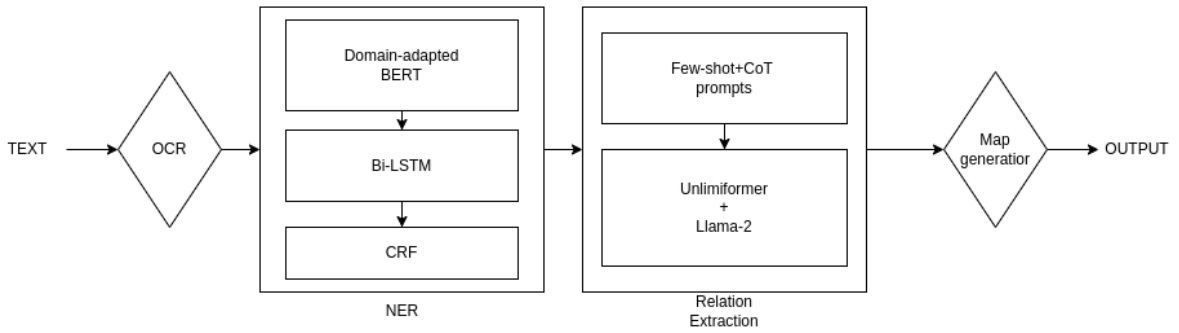


Figure 4.1 Processing pipeline for structured information extraction and mapping from historical texts. Input text is first digitized using OCR. Named Entity Recognition (NER) is performed using a domain-adapted BERT model followed by a Bi-LSTM and CRF layer. Relation extraction is guided by few-shot Chain-of-Thought (CoT) prompting with Unlimiformer and Llama-2. The extracted relations are then visualized via a map generation module.

In this section, we describe each step involved in building the pipeline, in the order in which they were carried out.

4.3.1 OCR and Text Extraction

The initial phase of our pipeline focused on converting scanned historical documents into machine-readable text. Many of these sources were manually digitised and existed only as image-based PDFs, posing significant challenges for standard Optical Character Recognition (OCR) tools. Several issues complicated the extraction process, including degraded print quality, irregular or antiquated typefaces, and non-standard, inconsistent layouts typical of historical publications. These factors often led to partial or inaccurate recognition of characters and words, particularly in dense or annotated sections.

We initially evaluated widely used open-source tools such as Tesseract [150] and PyPDF2 [151]. While both tools have proven effective in general OCR tasks, they struggled with the specific demands of our corpus due to the noisy nature of the scans. Blurred images, faded ink, and complex formatting meant that the tools frequently misread or omitted critical portions of text, making the raw output unsuitable for downstream NLP processing without substantial correction and cleaning.

To overcome these limitations, we adopted Google’s Document AI (GoogleOCR), a cloud-based OCR engine that leverages deep learning models and large scale language modeling to achieve state of the art accuracy [152]. It was especially effective at preserving layout structure, handling multilingual text, and correctly interpreting non-standard fonts—making it particularly suitable for our historical sources. Manual inspection of the OCR output confirmed that the extracted text was of sufficient quality for subsequent processing.

Following OCR, we applied a series of normalization steps to prepare the text for downstream NLP tasks. First, context aware spelling correction was used to address OCR-induced errors and standardize archaic variants. All text was then converted to UTF-8 to ensure character encoding consistency across the corpus. Next, we implemented a rule-based sentence segmentation system designed to accommodate historical punctuation, abbreviations, and irregular spacing. Finally, we developed a custom rule-based module to expand commonly used nineteenth century abbreviations (e.g., “Govt.” to “Government”) and to repair fragmented or fused words resulting from OCR inaccuracies.

4.3.2 Named Entity Recognition (NER)

Like we have mentioned, our NER pipeline was designed to extract three key categories of entities: **Geographical Entities**, **People**, and **Commodities**, which are essential for reconstructing trade networks from our historical corpus. To determine the most effective strategy of building our NER model, we trained and evaluated several state of the art NER models, each based on different NLP architectures.

Previous NER techniques

SpaCy NER: SpaCy’s built-in NER system is based on a transition-based parser and convolutional neural networks [153]. It predicts entities using an ensemble of dense representations and feature-based contextual cues, optimized for speed and efficiency. However, its performance typically lags behind transformer-based or sequence labeling architectures on domain-specific or noisy data.

BiLSTM (Bidirectional Long Short-Term Memory): BiLSTMs are a type of recurrent neural network (RNN) that read sequences in both forward and backward directions, capturing long range dependencies in text. They are widely used in sequence labeling tasks like NER because of their ability to model local context effectively [62, 63].

BERT (Bidirectional Encoder Representations from Transformers): BERT is a deep learning model based on the Transformer architecture. It is pretrained on large general corpora using a masked language modeling objective. Its key strength lies in contextual embeddings: each word’s representation is informed by both its left and right context. This makes BERT highly effective for downstream tasks such as NER when finetuned on task-specific data [66, 67].

CRF (Conditional Random Field): CRFs are probabilistic models used to predict structured outputs. In NER, they are typically used as the final layer in a model to ensure that the predicted sequence of labels is globally coherent (e.g., avoiding illegal sequences like an entity type beginning without a proper start tag) [61].

Combining these components, i.e. the contextual embeddings obtained from BERT, sequential understanding from BiLSTMs, and structured prediction using CRFs, has become a standard and powerful approach for high performance NER systems.

Domain-Adaptive Pretraining

As discussed in the literature review, general-purpose language models such as BERT are pretrained on modern corpora, typically consisting of news articles, Wikipedia entries, and web text [70]. As a result, they are ill suited for processing historical documents that feature archaic vocabulary, non-standard orthography, colonial-era toponyms, and syntactic conventions not commonly found in contemporary English [75, 76]. To address this mismatch, we adopted a domain-adaptive pretraining (DAPT) approach, wherein a pretrained language model is further trained on domain-specific text to better align it with the linguistic characteristics of the target corpus [87, 154, 155].

For this purpose, we curated a domain-specific corpus comprising 63 historical documents from the late nineteenth and early twentieth centuries. These included census reports, regional gazetteers, travelogues, and ethnographic accounts, all pertaining to the Western Himalayan region. After pre-processing to remove metadata and OCR artifacts, the resulting corpus amounted to approximately 1 million tokens. While relatively small by contemporary pretraining standards, this quantity is sufficient for effective domain adaptation, particularly when the base model already possesses general language understanding.³

We continued pretraining the original BERT-base model using the standard masked language modeling (MLM) and next sentence prediction (NSP) objectives. The MLM objective enables the model to learn domain-specific lexical and syntactic patterns by predicting randomly masked tokens within the input, while the NSP task helps in modeling the discourse-level coherence that is often present in narra-

³Link to the corpus used for domain-adaptive pretraining: <https://drive.google.com/drive/folders/15DzqjNgV3GUBTnVtWr2S1d7mJ-CA2okE?usp=sharing>

tive style historical texts. Pretraining was carried out over multiple epochs with a conservative learning rate to avoid catastrophic forgetting of previously learned representations.

This domain-adaptive pretraining step significantly enhanced the model’s ability to interpret the idiosyncrasies of historical language. For instance, the adapted model demonstrated improved recognition of historical place names (e.g., “Kulu,” “Chumba”), archaic terms for commodities, and role-specific references such as “Kardar” or “Shista Tapah.” This adapted language model formed the backbone for the downstream Named Entity Recognition task in our pipeline.

Model Architectures

We trained the following model architectures for Named Entity Recognition:

- **SpaCy Baseline:** The base Spacy model is finetuned on our annotated dataset using SpaCy’s NER pipeline. While computationally efficient and easy to deploy, it underperformed on longer and syntactically complex sentences, particularly for less frequent entity types.
- **BiLSTM+CRF:** A neural sequence labeling architecture that encodes input tokens with a bidirectional LSTM and uses a CRF layer for label decoding. This model captured local dependencies well but lacked the deep contextual understanding offered by transformer-based models.
- **BERT+BiLSTM+CRF:** This hybrid model used embeddings from a pretrained BERT model as input to a BiLSTM encoder, followed by a CRF decoder. By combining BERT’s contextual embeddings with BiLSTM’s sequential modeling and CRF’s structured prediction, the model achieved substantial performance gains over the earlier approaches.
- **Domain-Adapted BERT+BiLSTM+CRF:** This variant used the BERT model further pretrained on our domain-specific historical corpus. It consistently outperformed all other models, especially in disambiguating entities that share surface forms (e.g., place names that are also surnames or occupational titles).

Overall, the domain-adapted BERT+BiLSTM+CRF model emerged as the most effective architecture for our task, demonstrating the importance of tailoring language models to historical corpora with distinctive vocabulary and syntactic patterns. Detailed results are present in § 4.4.1.

4.3.3 Relation Extraction

Given the text and the entities present in it, the next step in the pipeline was to uncover trade-related relationships between places and commodities. To achieve this, we employed a prompt-based relation extraction strategy leveraging large language models (LLMs). Due to the lack of annotated relational datasets in our domain, traditional supervised approaches were infeasible. Instead, we adopted various learning paradigms using GPT-3.5 [156] and LLaMA-2 [157] as backend models, selected for their

strong generalization capabilities across a wide range of natural language understanding tasks without requiring extensive task-specific finetuning.

Our objective was to extract structured relational triples between place names and commodities, in the form `<Place> --trades--> <Commodity>`. Given an unstructured segment of historical prose, we first applied our Named Entity Recognition (NER) model to identify relevant entities. The extracted entities, along with the original text, were then passed to the relation extraction module, which inferred the semantic relationships between them. We experimented with three prompting strategies: zero-shot prompting, few-shot prompting, and few-shot prompting with chain-of-thought reasoning, since all three have been shown to be successful in past literature [158–160].

In the **zero-shot** setting, the model was provided with a plain task instruction to extract the relevant relations based on the text and the identified entities. While this is the most straightforward approach, it struggled with the complexity and ambiguity inherent in our sources—particularly when relationships were not explicitly stated or when relevant entities were dispersed across long spans of text.

In the **few-shot** setting, the prompt included a small number of representative input-output examples that illustrated how relational triples should be extracted from historical sentences. These examples contained a passage of text, the named entities present, and the correct relations among them. This approach significantly improved performance over the zero-shot baseline by grounding the model’s expectations and helping it generalize to structurally similar but previously unseen instances.

Our most effective approach was **few-shot prompting with chain-of-thought (CoT) reasoning**. In this setup, each example in the prompt included a historical sentence, its associated triple, and a short reasoning step that explained how the relation was inferred. This served two key purposes: first, it enhanced the interpretability of the model’s outputs; and second, it encouraged the model to decompose the task into smaller reasoning steps, thereby reducing errors caused by complex syntax, implicit references, or distant entity mentions. Theoretically, chain-of-thought prompting acts as a form of intermediate supervision embedded within the prompt itself. It guides the model through the latent inference steps required to extract implicit or pragmatically enriched relationships, an especially valuable property in historical texts, where relational cues are often indirect or dispersed.

To further improve contextual understanding, particularly in longer documents such as gazetteers and ethnographic reports, we integrated the Unlimiformer framework with LLaMA-2. This modification allows the model to process inputs of greater than 100,000 tokens, far exceeding the standard context window of typical transformer models. Extended context windows were crucial for maintaining coherence and enabling accurate relation extraction across paragraph-length segments, where relevant entities and their interactions might span multiple sentences or even entire sections.

In order to evaluate the performance of our relation extraction strategies, we constructed a gold-standard dataset in the absence of any pre-existing annotated corpus. Specifically, we randomly selected two documents from our collection of twenty two historical texts and manually annotated all trade-related relationships between place names and commodities within them. This curated subset served as the benchmark against which we assessed the outputs of our LLM-based models across the three

prompting strategies. Although this evaluation setup is limited in scale and may not capture the full linguistic diversity of the broader corpus, it nonetheless provides a meaningful and task-specific standard for comparison. Our results showed that the few-shot with chain-of-thought reasoning method consistently outperformed the other two, achieving higher precision and recall. These findings support our claim that prompting large language models with carefully crafted examples and intermediate reasoning steps is an effective and scalable strategy for extracting trade relations from unstructured historical text. Detailed results are present in § 4.4.2.

Together, these methods enabled a robust and scalable approach to relation extraction in a low resource, domain specific setting. The resulting structured data formed the foundation for downstream visualizations and analyses of commodity flows and place-people-commodity networks across the Western Himalayas.

4.3.4 Cartographic Mapping

Following relation extraction, we geolocated all identified place names using the GeoNames gazetteer as the primary reference. As mentioned before, the historical place names could differ in their spellings across documents. Hence, to disambiguate these place names, For each identified location, we performed a fuzzy matching procedure using the `rapidfuzz` library, which calculates string similarity based on Levenshtein distance. The coordinates corresponding to the best match were assigned to the entity. This output was further verified through manual review and performed reliably across the dataset.

The spatial data was visualized using a suite of Python libraries, including `GeoPandas`, `Folium`, and `Matplotlib`’s `Basemap` toolkit. Trade flows were represented as directional arcs between locations, with edge thickness denoting the frequency of co-occurrence or the reported trade volume (where quantifiable). Entities such as markets or trade hubs were encoded as nodes, with node size corresponding to their centrality in the extracted trade network.

We also generated period specific maps (e.g., early vs. late 19th century) to illustrate the temporal evolution of trade dynamics. This enabled us to observe changes in commodity routes, the emergence or decline of key trade centers, and shifts in regional connectivity across different phases of colonial and post-colonial economic restructuring.

4.4 Results

4.4.1 NER Performance

We evaluated four Named Entity Recognition (NER) models on our domain-specific annotated corpus, comparing their performance across precision, recall, and F1 score. As shown in Figure 4.2, the **domain-adapted BERT+BiLSTM+CRF** model (denoted as BERT+BiLSTM+CRF*) outperformed all other models across all three metrics, achieving a precision of 0.94, recall of 0.82, and an F1 score of 0.88.

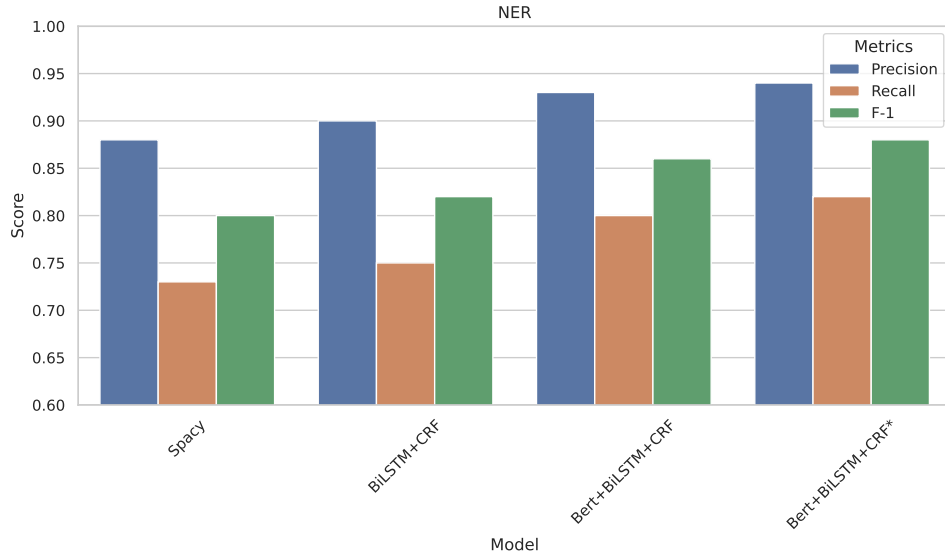


Figure 4.2 Comparison of NER model performance across three configurations: (1) SpaCy baseline, (2) BiLSTM + CRF, and (3) BERT + BiLSTM + CRF. The final model, marked with an asterisk, reflects the domain-adapted version used in our extraction pipeline. Metrics include Precision, Recall, and F1 score, with the domain-adapted BERT-based model showing the best overall performance.

The baseline **SpaCy** model achieved respectable precision (0.88) but significantly lower recall (0.73), resulting in an F1 score of 0.80. This reflects a tendency to be conservative in labeling, missing several valid entities, particularly those with archaic or region specific spellings. The **BiLSTM+CRF** model, though an improvement over SpaCy, still struggled with generalizing across the domain’s linguistic variety, particularly in cases involving long or compound entities. It achieved a precision of 0.90, recall of 0.75, and F1 of 0.82.

Adding BERT embeddings to the BiLSTM+CRF architecture (**BERT+BiLSTM+CRF**) resulted in a substantial performance boost, particularly in recall (0.80), as the model could better contextualize rare or syntactically embedded entities. This configuration achieved 0.93 precision and an F1 score of 0.86.

The best performance was observed in the domain-adapted BERT+BiLSTM+CRF model, which benefitted from pretraining on a 1 million token corpus of historical texts. The adaptation allowed the model to better recognize historical place names, commodity terms, and caste or occupation labels that are poorly represented in modern corpora. Notably, this model maintained high precision (0.94) while improving recall (0.82), indicating that it was not only accurate in its predictions but also able to identify a greater number of relevant entities.

Overall, these results highlight the value of domain adaptation and architectural complexity in building high performing NER systems for low resource, historically grounded corpora.

4.4.2 Relation Extraction Performance

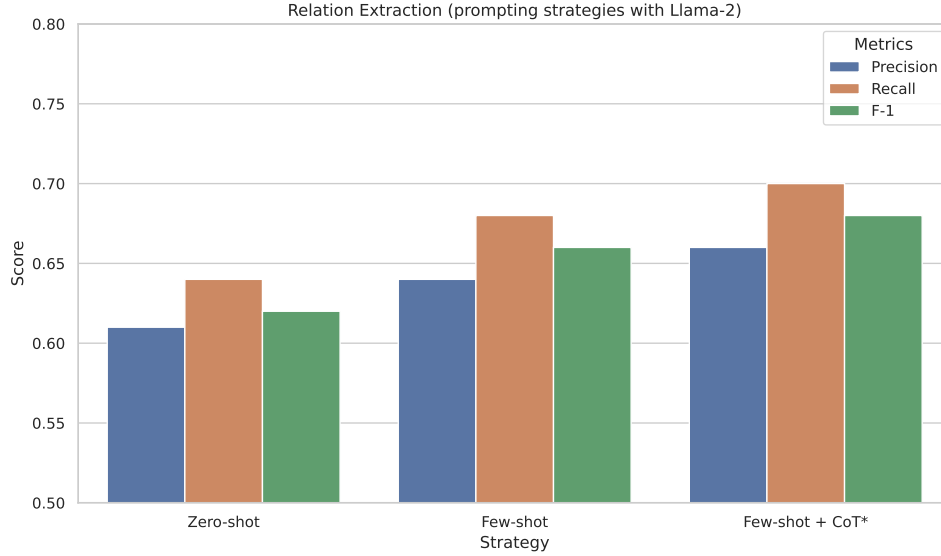


Figure 4.3 Performance comparison of prompting strategies for relation extraction using Llama-2. Zero-shot, few-shot, and few-shot with Chain-of-Thought (CoT) prompting are evaluated. The few-shot + CoT* strategy yields the best Precision, Recall, and F1 scores, demonstrating the value of structured reasoning in improving extraction quality for historical text.

We evaluated relation extraction performance through two complementary analyses: one focused on the effectiveness of different prompting strategies (Figure 4.3), and the other comparing model backends (Figure 4.4). In both analyses, we report precision, recall, and F1 scores based on a manually constructed gold-standard dataset comprising two annotated historical texts.

In the first analysis (Figure 4.3), we compared three prompting strategies: **zero-shot**, **few-shot**, and **few-shot with chain-of-thought reasoning**, using LLaMA-2 as the base model. Zero-shot prompting performed the weakest overall, achieving an F1 score of 0.62. Although it showed slightly higher recall than precision (0.64 vs. 0.61), its inability to disambiguate implicit or multi-hop relations led to frequent false positives and omissions. Few-shot prompting improved all metrics, with F1 rising to 0.66. The best results were obtained using **few-shot prompting with chain-of-thought reasoning**, which achieved a precision of 0.66, recall of 0.70, and an F1 score of 0.68. This supports our earlier hypothesis that intermediate reasoning steps help models better infer latent relationships, particularly in cases where relevant entities are syntactically distant or distributed across sentences.

In the second analysis (Figure 4.4), we compared the performance of three model architectures: **LLaMA-2**, **GPT-3.5**, and **Unlimiformer-extended LLaMA-2**. Among these, **GPT-3.5 with few-shot + CoT prompting** achieved the highest overall F1 score of **0.76**, with strong recall (0.81) and balanced precision (0.72). This suggests that GPT-3.5 not only captured more trade relationships but also main-

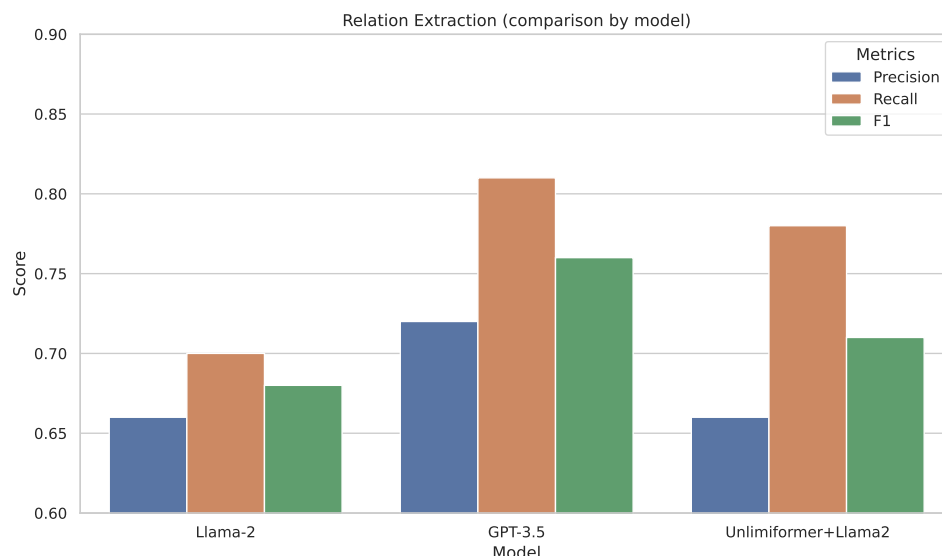


Figure 4.4 Comparison of relation extraction models on few-shot prompting tasks using Chain-of-Thought reasoning. Llama-2 and GPT-3.5 are evaluated against our final approach—Unlimiformer + Llama-2—which achieves the highest Precision, Recall, and F1 scores. This performance highlights the effectiveness of combining long-context handling with advanced instruction tuning for relation extraction in historical texts.

tained a relatively low false positive rate. Interestingly, while **Unlimiformer+LLaMA-2** did not match GPT-3.5 in overall F1 (0.71), it showed the highest recall (0.78) among all models, particularly benefiting from the extended context window of 65,000 tokens. This allowed it to better handle long-form ethnographic and administrative documents, where mentions of place names, people, and commodities may span paragraphs. The base **LLaMA-2** model performed the weakest, with an F1 of 0.68, limited by its shorter context length and lower capacity for abstract inference.

Overall, these results confirm the effectiveness of large language models for extracting historical trade relationships in a low-resource, domain-specific setting. They also highlight the benefits of prompt engineering, especially few-shot examples and chain-of-thought reasoning, as well as the utility of longer context windows in modeling complex narrative documents.

4.4.3 Visualization of Historical Trade

The extracted entities and relations were geocoded and visualized to uncover spatial and temporal patterns in historical trade across the Western Himalayas. These visualizations function as interpretable summaries of the structured outputs generated by our pipeline and offer an intuitive means to explore trade flows that would otherwise remain obscured in unstructured historical text.

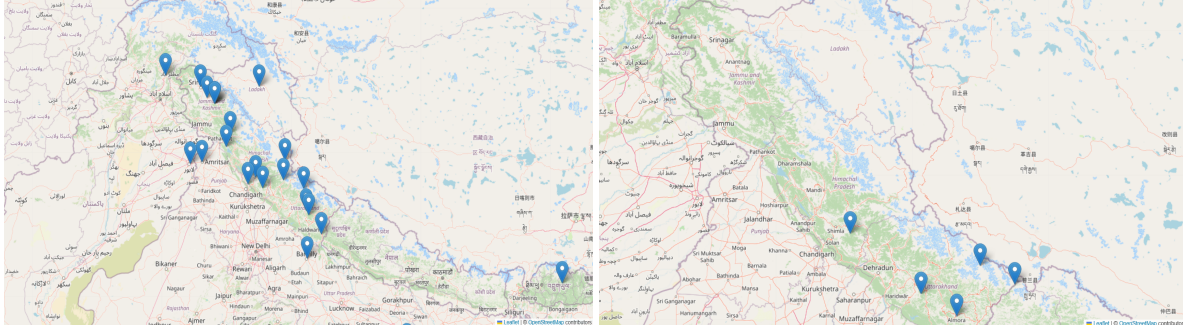


Figure 4.5 Comparison of cloth trade locations across two time periods. The map on the left depicts major locations involved in the trade of cloth at the beginning of the 19th century (circa 1810), while the right shows the situation by the end of the century (1890s). The decline of cloth (wool) trade in Kashmir across the 19th century can be clearly seen.

Figure 4.5 presents a comparative view of cloth trade networks at two different points in time—early 19th century (circa 1810) and late 19th century (1890s). Primary sources from each period were processed independently, and entity-relation triples related to cloth were extracted and mapped accordingly. The left-hand map displays a dense cluster of cloth-related trade activity centered in Kashmir, Punjab, and adjacent hill regions. This spatial distribution aligns with historical accounts of Kashmir’s centrality in the woollen cloth trade during the early colonial period. In contrast, the right-hand map, based on later sources, reveals a noticeable decline in trade nodes connected to Kashmir, with the spatial focus shifting southward and eastward into regions of present-day Himachal Pradesh and the northern plains.

This temporal shift in trade geography not only corresponds with broader economic and infrastructural changes during the late 19th century but also reinforces arguments from the historical literature regarding the decline of Kashmir’s dominance in regional textile trade (see Section 2.2.4).

More broadly, this case study demonstrates the practical utility of our computational pipeline as a historical research tool. By converting unstructured textual references into structured, mappable data, our approach enables historians to trace the evolution of commodity networks over time and space with greater clarity and scale than would be feasible through manual analysis alone. The ability to compare trade networks across different periods, visualize shifting centers of economic activity, and correlate extracted data with archival knowledge offers a novel, data-driven dimension to historical inquiry into Himalayan trade systems.

4.5 Discussion and Future Work

The computational pipeline developed in this study represents a significant advancement in digital historical methods, offering a scalable, reproducible approach for extracting and visualizing trade networks from historical texts. By integrating techniques from Natural Language Processing and spatial analysis, the framework addresses several longstanding limitations in the manual processing of historical corpora. Nonetheless, a number of challenges and areas for further development remain.

First, while we can identify which commodities are traded in which places, we are currently unable to extract the volumes of these commodities being traded, which would give us a better understanding of the amount of trade occurring. Additionally, future work can also focus on classifying which places serve as importers and which as exporters of these commodities to provide a more nuanced and in-depth understanding of the trade routes.

Second, the generalizability of the pipeline beyond the Western Himalayas remains an open question. The models were trained and evaluated on a geographically and temporally specific corpus; their applicability to other regions, periods, or genres of historical writing is yet to be assessed. To address this, future research could extend the training dataset to include historical texts from other parts of South and Central Asia. This would also require the development of multilingual and cross-script historical datasets, especially in Persian, Urdu, and regional Indic languages that are prevalent in the historical record.

Third, while the current system demonstrates methodological feasibility, its broader utility would be enhanced by the development of an open-source, user-friendly interface. We envision an interactive dashboard that enables historians to upload digitized texts and receive structured outputs such as annotated entities, extracted trade relations, and geospatial visualizations with options to filter results by time period, commodity, or region. Such a platform would make computational methods more accessible to scholars in the humanities, fostering interdisciplinary collaboration and wider adoption of digital tools.

4.6 Conclusion

This chapter has introduced an end-to-end computational framework for the extraction and visualization of historical trade networks, focusing on the Western Himalayas in the nineteenth and twentieth centuries. By combining OCR, domain-adapted Named Entity Recognition, prompt-based relation extraction, and geospatial mapping, the system translates unstructured historical narratives into structured, analyzable data.

The results demonstrate the value of computational methods in augmenting traditional historical inquiry. The pipeline successfully identified key entities and relations across a complex historical corpus, while the resulting visualizations uncovered both well-known and previously overlooked patterns in trade networks. These findings underscore the potential of digital tools to scale analysis, surface latent structures in text, and support new forms of historical interpretation.

Ultimately, this work lays the groundwork for future interdisciplinary research at the intersection of history and computer science. By making large-scale text analysis more feasible and interpretable, the approach contributes not only to the historiography of the Western Himalayas, but also to broader efforts to digitally reimagine the past.

Chapter 5

Conclusion

Our Work

This thesis set out to examine the interconnected histories of trade and pastoralism in the Western Himalayas during the nineteenth and twentieth centuries and to explore how computational tools, particularly those drawn from Natural Language Processing (NLP), could assist in the historical reconstruction of trade networks from unstructured textual sources. By combining archival research with computational methods, this thesis contributes to both historical scholarship and the growing field of computational humanities.

From a historical perspective, this thesis reframes the decline of pastoralism in the Western Himalayas not as a purely colonial imposition but as a more complex phenomenon shaped by both structural constraints and the agency of local communities. Through a close reading of gazetteers, census records, and ethnographic accounts, we highlight the intricate relational connections between people, places, and commodities that underpinned mountain economies. Transhumant pastoralist communities such as the Gaddis, Gujjars, Bhotiyas, and Changpas were not merely passive actors but active agents whose seasonal migrations sustained regional trade networks and facilitated the flow of essential commodities like wool, salt, and grain across diverse ecological zones.

While colonial policies like forest reservation, land settlement, and taxation undoubtedly disrupted these patterns of mobility and exchange, they did not operate in isolation. Pastoralist communities actively negotiated and responded to these structural changes. Many transitioned into settled agriculture, wage labor, and government employment, while others pursued hybrid strategies that combined traditional herding with new economic opportunities in education, local markets, and state institutions. These adaptive responses challenge the dominant narrative of passive victimhood and instead reveal strategic decision-making and social transformation from within. By foregrounding these dynamics, the thesis contributes to the historiography by illustrating how colonial interventions intersected with internal processes of adaptation and change, reshaping the relationships between people, commodities, and landscapes in the Western Himalayas.

From a computational standpoint, we manually annotate and build a dataset for NER for the Western Himalayan region and then introduce *PastPaths*, a domain-adapted NLP pipeline developed to extract

historical trade information from archival texts. The pipeline includes OCR and text preprocessing to handle noisy, scanned documents, Named Entity Recognition (NER) using a domain-adapted BERT model combined with a BiLSTM and CRF for extracting place and commodity names - the model is highly generalisable i.e. we can feed any text from the region to it and obtain its output, and relation extraction using few-shot Chain-of-Thought (CoT) prompting with LLaMA-2. Chaining these components enables us to identify and infer the commodities traded by specific places, even when such relations are only implicitly expressed in the text. Finally, a map generation module visualizes trade flows, enabling a spatial understanding of historical economic activity.

A central contribution of this work lies in its interdisciplinary methodology. Rather than applying computational tools in isolation, the thesis adopts a two-way integrative framework. Historical questions guided the design of the computational pipeline. For instance, the need to identify place-commodity pairs and trace their movement across time directly informed by our work in history, where we perform an analysis of the trade flows across the Western Himalayas. At the same time, the outputs of the NLP models provided new empirical insights that enriched historical interpretation, such as highlighting commodity flows that were difficult to reconstruct manually. This reciprocal relationship demonstrates how NLP and historical analysis can be brought into productive dialogue. Thus the thesis contributes to the field of computational humanities by offering novel insights into how the two disciplines can be combined for research purposes.

Limitations and Future Directions

While the thesis makes several contributions, it also has limitations that point toward future research opportunities. OCR errors and orthographic inconsistencies in the source texts remain a major bottleneck. Although the NER and relation extraction components performed well, improving text preprocessing and integrating OCR correction techniques tailored to historical documents would enhance the pipeline's accuracy.

Second, the current pipeline extracts place-commodity pairs but does not infer the directionality of trade. As part of future work, we propose extending the model to classify places as either importers or exporters based on textual cues. This enhancement would provide more nuanced visualizations and deeper insights into regional trade dynamics.

Third, the dataset used in this thesis was limited in size and scope. Expanding the corpus to include more diverse sources across different languages and regions would improve model generalizability. We also propose building open-access tools and annotated datasets to allow other researchers to extend and apply our pipeline to new historical questions and geographies.

Finally, the thesis calls for deeper institutional collaboration between computer science and the human sciences. Our work highlights the need for shared platforms where historical scholars and NLP practitioners can co-design models that are both technically robust and historically meaningful. Fu-

ture efforts should focus on developing such collaborative infrastructures, including joint annotation projects, interdisciplinary workshops, and hybrid research teams.

In conclusion, this thesis offers a dual contribution to historical research and computational methods by demonstrating how historical understanding and NLP can be mutually reinforcing. It reinterprets the decline of pastoralism through a lens of agency and adaptation, develops a novel NLP pipeline for historical trade extraction, and models a form of interdisciplinary engagement that advances both fields. While challenges remain, the research opens promising avenues for future work at the intersection of historical interpretation and computational analysis.

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